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FURTHER MATHEMATICS • UPPER SIXTH

UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 19

CONTINUOUS RANDOM VARIABLES

SUGGESTED TIME: 4 PERIODS

IN THIS CHAPTER

Ce chapitre introduit les variables aléatoires continues à travers leur fonction de densité, leur fonction de répartition, et les mesures de position et de dispersion associées. Il prépare l'élève à résoudre des problèmes concrets d'ingénierie et de sciences sociales.

- Définir et identifier une fonction de densité de probabilité (PDF) et vérifier ses propriétés.
- Calculer l'espérance, la variance et le mode d'une variable aléatoire continue.
- Déterminer la fonction de répartition (CDF) à partir de la PDF et réciproquement.
- Utiliser la CDF pour calculer des probabilités d'intervalles.
- Trouver la médiane, les quartiles et tout quantile d'une distribution continue.
- Résoudre des problèmes d'application mettant en jeu plusieurs notions du chapitre.

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COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

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CHAPTER OBJECTIVES

- Définir et identifier une fonction de densité de probabilité (PDF) et vérifier ses propriétés.
- Calculer l'espérance, la variance et le mode d'une variable aléatoire continue.
- Déterminer la fonction de répartition (CDF) à partir de la PDF et réciproquement.
- Utiliser la CDF pour calculer des probabilités d'intervalles.
- Trouver la médiane, les quartiles et tout quantile d'une distribution continue.
- Résoudre des problèmes d'application mettant en jeu plusieurs notions du chapitre.
- Interpréter graphiquement la PDF, la CDF et les mesures de position.
- Éviter les erreurs classiques : confusion PDF/CDF, omission de la normalisation, mauvaises bornes d'intégration.

PREREQUISITES

- Intégration définie et indéfinie (calcul d'aires, primitives).
- Notions de probabilités discrètes : espace probabilisé, espérance, variance.
- Fonctions usuelles : exponentielle, polynômes, fonctions par morceaux.
- Résolution d'équations et d'inéquations simples.
- Lecture et tracé de graphiques de fonctions.

1 FONCTION DE DENSITÉ DE PROBABILITÉ (PDF)

Dans les rues de Douala, le temps d'attente d'un taxi suit une loi continue : un passager peut attendre 2 minutes, 7.3 minutes, 12.87 minutes… n'importe quelle valeur réelle dans un intervalle. Un ingénieur veut modéliser cette attente pour optimiser le dispatching des véhicules. Comment décrire mathématiquement cette variable aléatoire continue ?

Le problème est le suivant : pour une variable aléatoire discrète (nombre de passagers, résultat d'un dé), on attribue une probabilité à chaque valeur possible. Mais le temps d'attente peut prendre une infinité non dénombrable de valeurs : la probabilité d'attendre exactement 5.000 … minutes est nulle ! Il faut donc un nouvel outil : au lieu de probabilités attachées à des points, on utilise une **densité de probabilité**, et les probabilités deviennent des aires sous une courbe. C'est l'objet de cette section.

1.1 De l'intuition à la définition

Observation. Imaginons que l'on mesure le temps d'attente de 10 000 passagers et que l'on trace un histogramme des fréquences avec des classes de plus en plus fines. À la limite, l'histogramme se lisse en une courbe $y = f(x)$, et la proportion d'attentes comprises entre a et b devient l'aire sous la courbe entre a et b. Cette courbe limite est la **fonction de densité de probabilité**.

DEFINITION **Variable aléatoire continue et fonction de densité**

Une variable aléatoire X est dite **continue** s'il existe une fonction $f : \mathbb{R} \to \mathbb{R}$, appelée **fonction de densité de probabilité** (PDF, probability density function), telle que pour tous réels $a \le b$:

$$P(a \le X \le b) = \int_a^b f(x)\,dx.$$

On dit que X suit la loi de densité f, et l'on note parfois $X \sim f$. L'ensemble des x où $f(x) > 0$ s'appelle le **support** de X.

PROPERTY **Propriétés fondamentales d'une PDF**

Une fonction f est une fonction de densité de probabilité si et seulement si :

1. Positivité : $f(x) \ge 0$ pour tout $x \in \mathbb{R}$;
2. Normalisation : $\int_{-\infty}^{+\infty} f(x)\,dx = 1$.

En conséquence, pour tout réel c : $P(X = c) = \int_c^c f(x)\,dx = 0$. Ainsi $P(a \le X \le b) = P(a < X < b)$: inclure ou exclure les bornes ne change rien pour une variable continue. (Propriété à connaître ; sa justification découle directement de la définition de l'intégrale.)

WARNING **La densité n'est pas une probabilité**

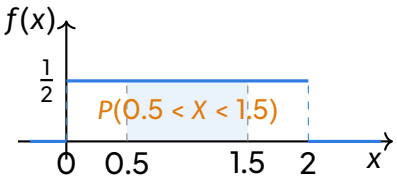
$f(x)$ n'est pas la probabilité que X vaille x. Une densité peut dépasser 1 ! Par exemple la loi uniforme sur $[0; 0.5]$ a pour densité $f(x) = 2$ sur cet intervalle. Seules les aires (intégrales) sont des probabilités, donc comprises entre 0 et 1. Dire « $f(3) = 0.7$ donc $P(X = 3) = 0.7$ » est une erreur classique et grave à l'examen.

1.2 Exemples classiques de lois continues

La loi uniforme. C'est le modèle le plus simple : la densité est constante sur un intervalle $[a; b]$ et nulle ailleurs. Pour que l'aire totale (un rectangle de base $b - a$) vaille 1, la hauteur doit être $\frac{1}{b-a}$:

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{si } a \le x \le b, \\ 0 & \text{sinon.} \end{cases}$$

On note $X \sim U[a; b]$. La probabilité $P(c \le X \le d)$ est simplement la longueur relative $\frac{d-c}{b-a}$ (pour $[c; d] \subset [a; b]$).

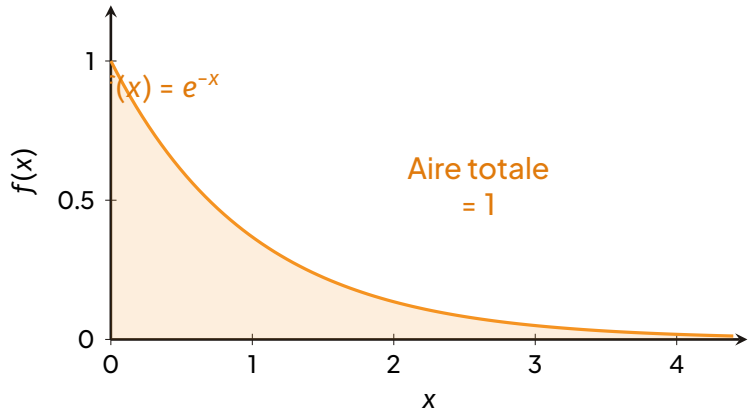


Densité de la loi uniforme sur $[0; 2]$; l'aire hachurée vaut $P(0.5 < X < 1.5) = 0.5$.

La loi exponentielle. Elle modélise les temps d'attente (durée de vie d'un composant, attente d'un taxi !). Pour un paramètre $\lambda > 0$:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{si } x \ge 0, \\ 0 & \text{si } x < 0. \end{cases}$$

Vérifions la normalisation : $\int_0^{+\infty} \lambda e^{-\lambda x}\, dx = \left[-e^{-\lambda x}\right]_0^{+\infty} = 0 - (-1) = 1$. La positivité est immédiate. C'est donc bien une PDF.



Densité exponentielle $f(x) = e^{-x}$ ($\lambda = 1$) : l'aire totale sous la courbe vaut 1.

La loi triangulaire. La densité a la forme d'un triangle sur $[a; b]$, nulle ailleurs, avec un sommet en $c \in [a, b]$. L'aire d'un triangle étant $\frac{1}{2} \times$ base $\times$ hauteur, la hauteur au sommet doit être $h = \frac{2}{b-a}$ pour que l'aire totale vaille 1. La densité s'écrit alors par morceaux :

$$f(x) = \begin{cases} \dfrac{2(x-a)}{(b-a)(c-a)} & \text{si } a \le x \le c, \\ \dfrac{2(b-x)}{(b-a)(b-c)} & \text{si } c \le x \le b, \\ 0 & \text{sinon.} \end{cases}$$

Cas particulier : triangle rectangle en a ($c = b$) : $f(x) = \frac{2(x-a)}{(b-a)^2}$ sur $[a, b]$; triangle rectangle en b ($c = a$) : $f(x) = \frac{2(b-x)}{(b-a)^2}$ sur $[a, b]$; triangle isocèle ($c = \frac{a+b}{2}$) : $f(x) = \frac{4}{b-a}\left(1 - \frac{2|x-c|}{b-a}\right)$.

EXEMPLE **Loi triangulaire sur $[0; 2]$ avec sommet en 2**

Ici $a = 0, b = 2, c = 2$ (triangle rectangle en b). La formule donne $f(x) = \frac{2(2-x)}{(2-0)(2-2)}$? Attention : le dénominateur $(b - c)$ s'annule. On utilise le cas particulier : $f(x) = \frac{2(b-x)}{(b-a)^2} = \frac{2(2-x)}{4} = 1 - \frac{x}{2}$ sur $[0, 2]$. Vérification : $\int_0^2 (1 - \frac{x}{2})dx = [x - \frac{x^2}{4}]_0^2 = 2 - 1 = 1$. C'est correct.

1.3 Détermination de la constante de normalisation

À l'examen, on donne souvent une densité dépendant d'une constante k et l'on demande de la déterminer.

METHOD **Trouver k pour que $f(x) = k\,g(x)$ soit une PDF**

1. Identifier le domaine D où $g(x) \neq 0$ (le support pressenti). Vérifier que $g(x) \geq 0$ sur D

(sinon aucun k ne convient).

2. Calculer l'intégrale $I = \int_D g(x)\,dx$ (souvent une intégrale sur un intervalle borné ou une intégrale impropre convergente).
3. Imposer la normalisation : $k \cdot I = 1$, d'où $k = \frac{1}{I}$ (nécessairement $k > 0$ car $I > 0$).
4. Conclure en écrivant la densité complète, avec ses deux morceaux ($f(x) = k\,g(x)$ sur D et 0 ailleurs).

WORKED EXAMPLE 1 **Détermination d'une constante**

Soit X une variable aléatoire continue de densité $f(x) = \begin{cases} kx(4 - x^2) & \text{si } 0 \le x \le 2, \\ 0 & \text{sinon.} \end{cases}$

Déterminer la valeur de k.

SOLUTION

Solution. Sur $[0;2]$, $x \ge 0$ et $4 - x^2 \ge 0$, donc $g(x) = x(4 - x^2) \ge 0$. La positivité de f impose $k > 0$.
Calculons l'intégrale de $g(x) = x(4 - x^2) = 4x - x^3$ sur $[0, 2]$:

$$\int_0^2 (4x - x^3)\,dx = \left[2x^2 - \frac{x^4}{4}\right]_0^2 = \left(2(4) - \frac{16}{4}\right) - 0 = 8 - 4 = 4.$$

La condition de normalisation $k \times 4 = 1$ donne $\boxed{k = \frac{1}{4}}$.

La densité est donc $f(x) = \frac{x(4 - x^2)}{4}$ sur $[0; 2]$, et 0 ailleurs.

1.4 Calcul de probabilités par intégration

Une fois la densité connue, toute probabilité s'obtient par intégration : c'est l'aire sous la courbe entre les bornes.

KEY POINTS **Formule fondamentale**

Pour une variable continue X de densité f :

$$P(a \le X \le b) = \int_a^b f(x)\,dx, \qquad P(X \ge a) = \int_a^{+\infty} f(x)\,dx, \qquad P(X \le b) = \int_{-\infty}^b f(x)\,dx.$$

Si f est nulle en dehors d'un intervalle $[m; M]$ (support borné), on restreint l'intégration à $[m; M] \cap [a; b]$.

EXEMPLE **Loi uniforme sur** $[0;2]$

Soit $X \sim U[0;2]$, de densité $f(x) = \frac{1}{2}$ sur $[0;2]$. Alors

$$P(0.5 < X < 1.5) = \int_{0.5}^{1.5} \frac{1}{2}\,dx = \frac{1}{2}(1.5 - 0.5) = 0.5.$$

C'est exactement l'aire du rectangle hachuré sur la première figure.

WORKED EXAMPLE 2 **Temps d'attente d'un taxi à Douala**

Le temps d'attente X (en minutes) d'un taxi est modélisé par une loi exponentielle de paramètre $\lambda = 0.2$: $f(x) = 0.2\,e^{-0.2x}$ pour $x \geq 0$.
Calculer la probabilité qu'un passager attende entre 5 et 10 minutes, puis plus de 15 minutes.

SOLUTION

Solution.

$$P(5 \leq X \leq 10) = \int_{5}^{10} 0.2\,e^{-0.2x}\,dx = \left[-e^{-0.2x}\right]_{5}^{10}$$
$$= -e^{-2} + e^{-1} = e^{-1} - e^{-2} \approx 0.3679 - 0.1353 \approx 0.233.$$

Pour l'attente supérieure à 15 minutes :

$$P(X \geq 15) = \int_{15}^{+\infty} 0.2\,e^{-0.2x}\,dx = \left[-e^{-0.2x}\right]_{15}^{+\infty} = 0 - (-e^{-3}) = e^{-3} \approx 0.0498.$$

Il y a donc environ 23.3% de chances d'attendre entre 5 et 10 minutes, et seulement 5% de chances d'attendre plus d'un quart d'heure.

WARNING **Erreurs fréquentes**

- Oublier le domaine : si f est définie sur $[0;2]$ seulement, alors $P(X \leq 5) = 1$ (et non $\int_0^5 f$ qui n'a pas de sens au-delà de 2).
- Intégrer sur les mauvaises bornes : $P(X \geq a)$ s'intègre de a à $+\infty$, pas de 0 à a.
- Confondre densité et probabilité : $f(x)$ peut valoir 2 ou 10 ; seule l'aire totale est contrainte à 1.
- Croire que $P(X = a) > 0$: pour toute variable continue, $P(X = a) = 0$.

DID YOU KNOW? **Pourquoi « densité » ?**

Le vocabulaire vient de la physique : comme la masse d'une tige non homogène est l'intégrale de sa densité linéique, la probabilité d'un intervalle est l'intégrale de la densité de probabilité. La probabilité totale, comme la masse totale, vaut 1. Les lois exponentielles sont utilisées chaque jour par les ingénieurs télécoms camerounais pour modéliser les durées entre appels dans un réseau !

EXERCISE 1 **À vous de jouer**

Soit $f(x) = \begin{cases} \dfrac{k}{x^2} & \text{si } x \geq 1, \\ 0 & \text{sinon.} \end{cases}$

1. Déterminer k pour que f soit une densité de probabilité.
2. Calculer $P(1 \leq X \leq 3)$ et $P(X \geq 2)$.

SOLUTION **Exercice 1**

1. $\displaystyle\int_1^{+\infty} \frac{k}{x^2}\,dx = k\left[-\frac{1}{x}\right]_1^{+\infty} = k(0+1) = k$. La normalisation impose $k = 1$.
2. $P(1 \leq X \leq 3) = \displaystyle\int_1^3 \frac{dx}{x^2} = \left[-\frac{1}{x}\right]_1^3 = 1 - \frac{1}{3} = \frac{2}{3}$; $\quad P(X \geq 2) = \displaystyle\int_2^{+\infty} \frac{dx}{x^2} = \frac{1}{2}$.

KEY POINTS **L'essentiel de la section**

- Une variable aléatoire continue est décrite par une **densité** $f \geq 0$ dont l'intégrale totale vaut 1.
- Les probabilités sont des **aires** : $P(a \leq X \leq b) = \displaystyle\int_a^b f(x)\,dx$, et $P(X = c) = 0$.
- La constante de normalisation k s'obtient en résolvant $k \displaystyle\int g(x)\,dx = 1$.
- Lois de référence : uniforme (f constante), exponentielle ($f(x) = \lambda e^{-\lambda x}$, $x \geq 0$), triangulaire.

2 ESPÉRANCE, VARIANCE ET MODE

Dans la section précédente, nous avons appris qu'une variable aléatoire continue X est complètement décrite par sa **fonction de densité de probabilité** (PDF) f : les probabilités sont des aires sous la courbe, $\Pr(a \le X \le b) = \int_a^b f(x)\,dx$, et l'aire totale vaut 1. Mais une courbe entière représente beaucoup d'information. En pratique, on cherche souvent quelques nombres qui résument la distribution : où est-elle centrée ? Quelle est sa dispersion ? Où se situe le pic ? Ces trois questions mènent à l'**espérance**, à la **variance** et au **mode** de X.

2.1 L'espérance $E(X)$

Du cas discret au cas continu.

Pour une variable aléatoire discrète prenant les valeurs $x_1, x_2, \dots$ avec les probabilités $p_1, p_2, \dots$, l'espérance est la moyenne pondérée

$$E(X) = \sum_i x_i\, p_i.$$

Pour une variable continue, la probabilité de tomber dans un minuscule intervalle $[x, x + dx]$ est approximativement $f(x)\,dx$. En remplaçant la somme par une intégrale (l'analogue continu de la sommation), on obtient la définition naturelle.

DEFINITION **Espérance d'une variable aléatoire continue**

Soit X une variable aléatoire continue de fonction de densité f définie sur son support S. L'**espérance** (ou **moyenne**) de X est

$$E(X) = \mu = \int_S x\, f(x)\,dx,$$

pourvu que l'intégrale converge. Si le support est $\mathbb{R}$ tout entier, $E(X) = \int_{-\infty}^{+\infty} x f(x)\,dx$.

Interprétation physique : le centre de gravité.

Imaginez la région sous la courbe de densité découpée dans une plaque mince d'épaisseur uniforme. Comme l'aire totale est 1, la plaque a une masse 1. L'espérance μ est exactement l'abscisse du **centre de gravité** de cette plaque : si vous placez un couteau sous la droite $x = \mu$, la plaque reste en équilibre. Une densité symétrique est donc centrée sur son axe de symétrie — un raccourci très utile.

PROPERTY **Linéarité de l'espérance**

Pour toutes constantes $a, b \in \mathbb{R}$ et toutes variables aléatoires continues X, Y :

$$E(aX + b) = aE(X) + b, \qquad E(X + Y) = E(X) + E(Y).$$

Justification. Par linéarité de l'intégrale,

$$E(aX+b) = \int_S (ax+b)f(x)\,dx = a\int_S xf(x)\,dx + b\int_S f(x)\,dx = aE(X)+b,$$

puisque $\int_S f(x)\,dx = 1$. ■

EXEMPLE **Moyenne d'une loi uniforme**

Soit $X \sim U(a,b)$, donc $f(x) = \dfrac{1}{b-a}$ pour $a \le x \le b$. Alors

$$E(X) = \int_a^b \frac{x}{b-a}\,dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}.$$

La moyenne est le milieu du support — exactement ce que la symétrie prévoit.

EXEMPLE **Moyenne d'une loi exponentielle**

Soit X de densité $f(x) = \lambda e^{-\lambda x}$ pour $x \ge 0$ ($\lambda > 0$). En intégrant par parties avec $u = x$, $dv = \lambda e^{-\lambda x}dx$:

$$E(X) = \int_0^{+\infty} x\lambda e^{-\lambda x}\,dx = \left[-xe^{-\lambda x}\right]_0^{+\infty} + \int_0^{+\infty} e^{-\lambda x}\,dx = 0 + \frac{1}{\lambda} = \frac{1}{\lambda}.$$

Ainsi, si un composant tombe en panne avec un taux $\lambda = 0,5$ par an, sa durée de vie moyenne est $1/0,5 = 2$ ans.

2.2 La variance Var(X)

Pourquoi la moyenne ne suffit pas.

Deux distributions de temps d'attente peuvent avoir toutes deux une moyenne de 10 minutes, l'une étant concentrée entre 9 et 11 minutes, l'autre s'étalant de 0 à 40 minutes. Il nous faut une mesure de la **dispersion** autour de la moyenne. Faire la moyenne de l'écart $X-\mu$ est inutile car les écarts positifs et négatifs s'annulent : $E(X-\mu) = E(X)-\mu = 0$. Faire la moyenne de l'écart au carré résout le problème.

DEFINITION **Variance et écart type**

La **variance** d'une variable aléatoire continue X de moyenne μ est

$$\mathrm{Var}(X) = E\left[(X-\mu)^2\right] = \int_S (x-\mu)^2 f(x)\,dx.$$

L'**écart type** est $\sigma = \sqrt{\mathrm{Var}(X)}$; il a la même unité que X.

PROPERTY **Formule de calcul de la variance**

$$\mathrm{Var}(X) = E(X^2) - \big[E(X)\big]^2 = E(X^2) - \mu^2, \qquad \text{où } E(X^2) = \int_S x^2 f(x)\,dx.$$

La démonstration est exigible à l'examen et très instructive.

Preuve. On développe le carré et on utilise la linéarité de l'espérance :

$$\begin{aligned}\mathrm{Var}(X) &= E\big[(X-\mu)^2\big] = E\big[X^2 - 2\mu X + \mu^2\big] \\ &= E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2. \quad \blacksquare\end{aligned}$$

WARNING **Une erreur très fréquente**

$E(X^2)$ n'est pas $\big[E(X)\big]^2$. Comme $\mathrm{Var}(X) \geq 0$, on a toujours $E(X^2) \geq \mu^2$, avec égalité seulement pour une variable constante. Calculez toujours $E(X^2)$ par sa propre intégrale $\int x^2 f(x)\,dx$.

PROPERTY **Effet d'une transformation affine sur la variance**

Pour des constantes $a, b \in \mathbb{R}$:

$$\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X).$$

L'ajout d'une constante b décale la distribution sans changer sa dispersion ; la multiplication par a multiplie la variance par a^2 (et l'écart type par $|a|$).

Preuve. La moyenne de $aX + b$ est $a\mu + b$, donc

$$\mathrm{Var}(aX+b) = E\Big[\big((aX+b) - (a\mu+b)\big)^2\Big] = E\big[a^2(X-\mu)^2\big] = a^2\mathrm{Var}(X). \quad \blacksquare$$

WORKED EXAMPLE 3 **Variance de la loi uniforme**

Soit $X \sim U(a, b)$. Montrer que $\mathrm{Var}(X) = \dfrac{(b-a)^2}{12}$.

SOLUTION

On sait que $\mu = \dfrac{a+b}{2}$. On calcule $E(X^2)$:

$$E(X^2) = \int_a^b \frac{x^2}{b-a}\,dx = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}.$$

Alors

$$\mathrm{Var}(X) = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{4(a^2+ab+b^2) - 3(a^2+2ab+b^2)}{12}$$
$$= \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12}. \quad \blacksquare$$

Vérification numérique : pour $U(0,12)$, $\mathrm{Var}(X) = 12$, $\sigma = 2\sqrt{3} \approx 3,46$.

WORKED EXAMPLE 4 **Moyenne et variance d'une loi exponentielle**

Soit $f(x) = \lambda e^{-\lambda x}$ pour $x \geq 0$. Trouver $\mathrm{Var}(X)$.

SOLUTION

On a déjà $\mu = 1/\lambda$. Pour $E(X^2)$, on intègre par parties deux fois (ou on utilise la formule de récurrence $I_n = \frac{n}{\lambda} I_{n-1}$ avec $I_n = \int_0^{+\infty} x^n \lambda e^{-\lambda x} dx$) :

$$E(X^2) = \int_0^{+\infty} x^2 \lambda e^{-\lambda x}\, dx = \Big[-x^2 e^{-\lambda x} \Big]_0^{+\infty} + \frac{2}{\lambda} \int_0^{+\infty} x \lambda e^{-\lambda x}\, dx = 0 + \frac{2}{\lambda} \cdot \frac{1}{\lambda} = \frac{2}{\lambda^2}.$$

Donc

$$\mathrm{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}, \qquad \sigma = \frac{1}{\lambda}.$$

Remarque remarquable : pour la loi exponentielle, l'écart type égale la moyenne.

2.3 Le mode

DEFINITION **Mode d'une variable aléatoire continue**

Le **mode** de X est la valeur de x en laquelle la densité f atteint son maximum. C'est la « région la plus probable » des valeurs — le sommet de la courbe de densité.

WARNING **Le mode n'est pas nécessairement unique**

Pour une distribution uniforme $U(a,b)$, la densité est constante sur $[a,b]$, donc chaque point de $[a,b]$ maximise f : le mode n'est pas unique. Plus généralement, une distribution peut avoir plusieurs modes (bimodale, …). Attention aussi : le mode n'est pas la valeur de f au sommet — c'est la valeur de x où le sommet se produit.

METHOD **Recherche du mode**

1. Identifier le support S de f.
2. Calculer $f'(x)$ et résoudre $f'(x) = 0$ à l'intérieur de S.
3. Étudier la nature de chaque point stationnaire (signe de f' ou dérivée seconde) et comparer avec les valeurs de f aux bornes de S : le maximum peut se trouver en un bord (ex. loi exponentielle, maximale en $x = 0$).
4. Conclure : le mode est la valeur de x donnant la plus grande valeur de f.

WORKED EXAMPLE 5 **Mode, moyenne et variance d'une loi triangulaire**

Un commerçant de Bamenda modélise la demande journalière X (en sacs de gari) par

$$f(x) = \begin{cases} \dfrac{x}{16} & 0 \le x \le 4, \\ \dfrac{8-x}{16} & 4 \le x \le 8, \\ 0 & \text{sinon.} \end{cases}$$

Trouver le mode, la moyenne et la variance de X.

SOLUTION

Vérification que f est une densité.

$$\int_0^4 \frac{x}{16}\,dx + \int_4^8 \frac{8-x}{16}\,dx = \left[\frac{x^2}{32}\right]_0^4 + \left[\frac{8x - x^2/2}{16}\right]_4^8 = \frac{16}{32} + \frac{(64-32)-(32-8)}{16} = \frac{1}{2} + \frac{1}{2} = 1.$$

Mode. Sur $[0,4]$, $f'(x) = \frac{1}{16} > 0$ (croissante) ; sur $[4,8]$, $f'(x) = -\frac{1}{16} < 0$ (décroissante). Le maximum est à la jonction : mode = 4, avec $f(4) = \frac{1}{4}$.
Moyenne. La densité est symétrique par rapport à $x = 4$, donc $\mu = 4$. Vérification par intégration :

$$E(X) = \int_0^4 \frac{x^2}{16}\,dx + \int_4^8 \frac{x(8-x)}{16}\,dx = \frac{64}{48} + \frac{1}{16}\left[4x^2 - \frac{x^3}{3}\right]_4^8 = \frac{4}{3} + \frac{1}{16}\left(\frac{256}{3} - \frac{128}{3}\right) = \frac{4}{3} + \frac{128}{48} = \frac{4}{3} + \frac{8}{3} = 4.$$

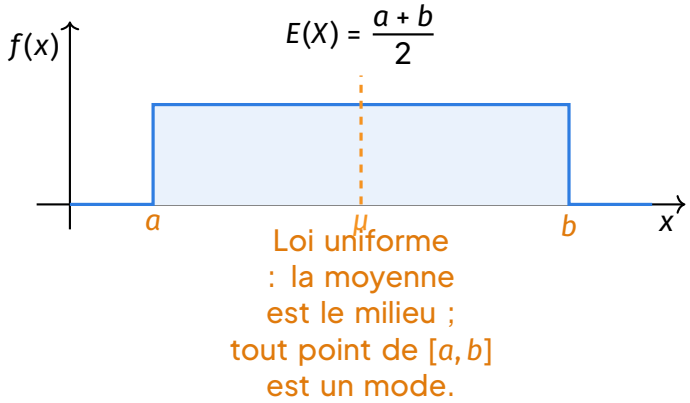
Variance.

$$E(X^2) = \int_0^4 \frac{x^3}{16}\,dx + \int_4^8 \frac{x^2(8-x)}{16}\,dx = \frac{256}{64} + \frac{1}{16}\left[\frac{8x^3}{3} - \frac{x^4}{4}\right]_4^8 = 4 + \frac{1}{16}\left(\frac{4096}{3} - 1024 - \frac{512}{3} + 64\right)$$

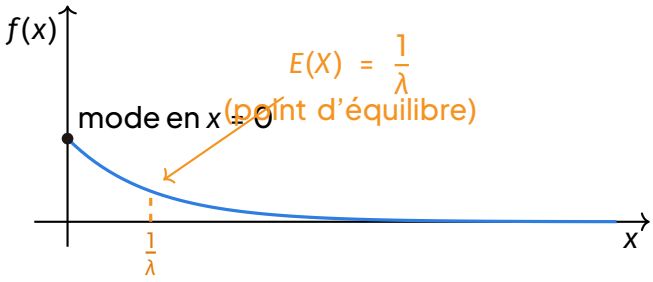
$$= 4 + \frac{1}{16}\left(\frac{3584}{3} - 960\right) = 4 + \frac{1}{16}\cdot\frac{704}{3} = 4 + \frac{44}{3} = \frac{56}{3}.$$

$$\mathrm{Var}(X) = \frac{56}{3} - 16 = \frac{8}{3}, \qquad \sigma = \sqrt{\tfrac{8}{3}} \approx 1{,}63 \text{ sacs}.$$

2.4 Interprétation graphique des trois mesures



PDF de la loi uniforme $U(a,b)$: l'espérance (trait pointillé) est le centre de gravité ; le mode n'est pas unique.



PDF de la loi exponentielle $f(x) = \lambda e^{-\lambda x}$ (ici $\lambda = 1$) : mode à la borne $x = 0$, moyenne en $x = 1/\lambda$.

DID YOU KNOW? **Origine du terme « espérance »**

Le concept d'espérance mathématique est né au XVIIe siècle avec le « problème des partis » : deux joueurs doivent interrompre une partie ; comment partager la mise équitablement ? Christiaan Huygens (1657) a formalisé l'idée de « valeur attendue » comme somme des gains pondérés par leurs probabilités. En français, on a gardé le mot « espérance » (au sens d'« espoir mathématique »), tandis que l'anglais utilise expectation. Au Cameroun, les compagnies d'assurance utilisent l'espérance pour calculer les primes : la prime « pure » est l'espérance du sinistre à indemniser.

EXPERIMENT **Simulation de la loi des grands nombres**

Si vous avez accès à un tableur (Excel, LibreOffice Calc) ou à Python :

1. Générez $N = 1000$ nombres aléatoires uniformes sur $[0, 1]$ (`ALEA()` ou `random.random()`).
2. Calculez la moyenne empirique $\bar{x}_N = \frac{1}{N}\sum x_i$.
3. Augmentez N progressivement ($10, 100, 1000, 10000$) et observez $\bar{x}_N$ converger vers la moyenne théorique $\mu = 0,5$.
4. Répétez avec une loi exponentielle (transformez U par $X = -\frac{1}{\lambda}\ln U$) et vérifiez la convergence vers $1/\lambda$.

Cette expérience illustre le théorème de la loi des grands nombres : la moyenne d'échantillon converge vers l'espérance.

KEY POINTS **Formules clés de cette section**

- Espérance : $E(X) = \int_S x f(x)\,dx$ — centre de gravité de la courbe de densité.
- Variance : $\mathrm{Var}(X) = E(X^2) - \mu^2$ avec $E(X^2) = \int_S x^2 f(x)\,dx$; $\sigma = \sqrt{\mathrm{Var}(X)}$.
- Linéarité : $E(aX + b) = aE(X) + b$; $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$.
- Mode : valeur(s) de x maximisant f ; résoudre $f'(x) = 0$ et vérifier les bornes.
- Résultats standards :
  - $U(a, b)$: $\mu = \frac{a+b}{2}$, $\mathrm{Var} = \frac{(b-a)^2}{12}$, mode non unique.
  - Exponentielle(λ) : $\mu = \sigma = \frac{1}{\lambda}$, $\mathrm{Var} = \frac{1}{\lambda^2}$, mode $= 0$.

EXERCISE 2 **Entraînement**

1. Soit $f(x) = \dfrac{3x^2}{8}$ pour $0 \le x \le 2$ (et 0 ailleurs). Calculer $E(X)$, $\mathrm{Var}(X)$ et le mode de X.
2. Pour $X \sim U(2, 10)$, donner $E(X)$ et $\mathrm{Var}(X)$ sans intégrer, puis trouver $E(3X - 1)$ et $\mathrm{Var}(3X - 1)$.
3. Une durée de vie T suit une loi exponentielle de moyenne 4 ans. Donner λ, $\mathrm{Var}(T)$ et le mode.
4. Approfondissement. Soit X de densité $f(x) = kx(4 - x)$ sur $[0, 4]$.
   (a) Déterminer k.

(b) Calculer $E(X)$ et $\mathrm{Var}(X)$.

(c) Trouver le mode de X.

SOLUTION **Exercice 1**

1. $E(X) = \int_0^2 \frac{3x^3}{8}dx = \frac{3}{8}\cdot\frac{16}{4} = \frac{3}{2}$. $E(X^2) = \int_0^2 \frac{3x^4}{8}dx = \frac{3}{8}\cdot\frac{32}{5} = \frac{12}{5}$. $\mathrm{Var}(X) = \frac{12}{5} - \frac{9}{4} = \frac{48-45}{20} = \frac{3}{20}$. Comme f est strictement croissante sur $[0,2]$, le mode est à la borne : $x = 2$.
2. $\mu = \frac{2+10}{2} = 6$, $\mathrm{Var}(X) = \frac{64}{12} = \frac{16}{3}$. $E(3X-1) = 3(6) - 1 = 17$; $\mathrm{Var}(3X-1) = 9\cdot\frac{16}{3} = 48$.
3. $\mu = \frac{1}{\lambda} = 4 \Rightarrow \lambda = \frac{1}{4}$; $\mathrm{Var}(T) = \frac{1}{\lambda^2} = 16$ ans^2 ; mode $= 0$.
4. a) $\int_0^4 kx(4-x)dx = k\left[2x^2 - \frac{x^3}{3}\right]_0^4 = k\left(32 - \frac{64}{3}\right) = k\frac{32}{3} = 1 \Rightarrow k = \frac{3}{32}$. b) $E(X) = \frac{3}{32}\int_0^4 x^2(4-x)dx = \frac{3}{32}\left[\frac{4x^3}{3} - \frac{x^4}{4}\right]_0^4 = \frac{3}{32}\left(\frac{256}{3} - 64\right) = \frac{3}{32}\cdot\frac{64}{3} = 2$. $E(X^2) = \frac{3}{32}\int_0^4 x^3(4-x)dx = \frac{3}{32}\left[x^4 - \frac{x^5}{5}\right]_0^4 = \frac{3}{32}\left(256 - \frac{1024}{5}\right) = \frac{3}{32}\cdot\frac{256}{5} = \frac{24}{5} = 4,8$. $\mathrm{Var}(X) = 4,8 - 4 = 0,8 = \frac{4}{5}$. c) $f(x) = \frac{3}{32}(4x - x^2)$, $f'(x) = \frac{3}{32}(4-2x) = 0 \Rightarrow x = 2$. $f''(2) < 0$, c'est un maximum. Mode $= 2$ (au centre par symétrie).

3 FONCTION DE RÉPARTITION (CDF)

In the previous section we learned how to summarise a continuous distribution by its expectation, variance and mode. These are single numbers that capture the centre, spread and peak of the distribution. However, there are many situations where we need the probability that the random variable falls below a certain threshold, or between two values. For example, an engineer designing a bridge in Douala might need the probability that the maximum daily load exceeds a safety limit, or a quality control manager at a cocoa processing plant in Kribi might want the probability that the moisture content of a batch lies between 6% and 8%. The tool that answers all such questions in one unified way is the **cumulative distribution function** (CDF).

3.1 Definition and Intuition

For a discrete random variable, the cumulative distribution function is a step function that jumps at each possible value. For a continuous random variable, the CDF is a smooth (or at least continuous) curve obtained by accumulating the area under the probability density function (PDF) from left to right.

DEFINITION **Cumulative Distribution Function (CDF)**

Let X be a continuous random variable with probability density function $f(x)$. The cumu-

lative distribution function (CDF), denoted by $F(x)$, is defined for every real number x by

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt.$$

- The variable of integration is a dummy variable; we use t to avoid confusion with the upper limit x.
- $F(x)$ represents the total area under the curve $y = f(t)$ from $-\infty$ up to x.
- Because $f(t) \ge 0$ everywhere, $F(x)$ is a **non-decreasing** function of x.
- The CDF is defined for **all** real x, not only on the support of X.

EXEMPLE **CDF of a Simple Triangular Distribution**

Let X have the triangular PDF

$$f(x) = \begin{cases} 2x, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find $F(x)$ for all real x.
Solution: We consider three regions.

- For $x < 0$: $f(t) = 0$ on $(-\infty, x]$, so $F(x) = 0$.
- For $0 \le x \le 1$:

$$F(x) = \int_{-\infty}^{x} f(t)\,dt = \int_{0}^{x} 2t\,dt = \left[t^2\right]_0^x = x^2.$$

- For $x > 1$: The total area under the PDF is 1, so $F(x) = 1$.

Hence

$$F(x) = \begin{cases} 0, & x < 0, \\ x^2, & 0 \le x \le 1, \\ 1, & x > 1. \end{cases}$$

3.2 Properties of the CDF

The CDF of any continuous random variable satisfies the following fundamental properties.

PROPERTY **Properties of the CDF**

Let X be a continuous random variable with CDF $F(x)$. Then:

1. Limits: $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to +\infty} F(x) = 1$.
2. Monotonicity: $F(x)$ is non-decreasing: if $x_1 < x_2$ then $F(x_1) \le F(x_2)$.

3. Continuity: $F(x)$ is continuous for all real x (in fact, it is absolutely continuous).
4. Range: $0 \le F(x) \le 1$ for all x.
5. Probability of an interval: For any $a < b$,

$$P(a < X \le b) = F(b) - F(a).$$

Because $P(X = a) = 0$ for a continuous variable, this also equals $P(a \le X \le b) = P(a \le X < b) = P(a < X < b)$.

WARNING **Common Mistake: Confusing PDF and CDF**

Never write $P(X = x) = f(x)$. For a continuous random variable, the probability of taking any exact value is zero: $P(X = x) = 0$. The PDF $f(x)$ is a density, not a probability. The CDF $F(x)$ gives actual probabilities: $F(x) = P(X \le x)$.

3.3 Relationship between PDF and CDF

The PDF and CDF are two sides of the same coin. The Fundamental Theorem of Calculus gives the precise link.

PROPERTY **Fundamental Relation: $f(x) = F'(x)$**

If F is differentiable at x, then

$$f(x) = \frac{d}{dx}F(x) = F'(x).$$

Conversely,

$$F(x) = \int_{-\infty}^{x} f(t)\,dt.$$

This holds at every point where f is continuous. At points where f has a jump discontinuity (e.g. at the boundaries of a uniform distribution), F is still continuous but not differentiable; the relation $f(x) = F'(x)$ holds almost everywhere.

METHOD **Moving between PDF and CDF**

1. Given $f(x)$, find $F(x)$: Integrate $f(t)$ from the lower bound of the support up to x. Treat the cases x below the support, inside the support, and above the support separately. Check that $F(x) \to 0$ as $x \to -\infty$ and $F(x) \to 1$ as $x \to +\infty$.
2. Given $F(x)$, find $f(x)$: Differentiate $F(x)$ on each interval where it is differentiable. The

support of f is where F is strictly increasing.

3. Check: The area under f must be 1, and F must be continuous, non-decreasing, with limits 0 and 1.

3.4 Computing the CDF for Common Distributions

We now derive the CDF for the three standard continuous distributions in the syllabus.

EXEMPLE **CDF of the Uniform Distribution**

Let $X \sim U(a, b)$ with PDF $f(x) = \frac{1}{b-a}$ for $a \le x \le b$, zero elsewhere.

$$F(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}$$

Derivation: For $x < a$, $F(x) = 0$. For $a \le x \le b$,

$$F(x) = \int_a^x \frac{1}{b-a}\,dt = \frac{x-a}{b-a}.$$

For $x > b$, $F(x) = 1$. The graph is a straight line segment from $(a, 0)$ to $(b, 1)$.

EXEMPLE **CDF of the Exponential Distribution**

Let $X \sim \text{Exp}(\lambda)$ with PDF $f(x) = \lambda e^{-\lambda x}$ for $x \ge 0$, zero elsewhere ($\lambda > 0$).

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \ge 0. \end{cases}$$

Derivation: For $x < 0$, $F(x) = 0$. For $x \ge 0$,

$$F(x) = \int_0^x \lambda e^{-\lambda t}\,dt = \left[-e^{-\lambda t}\right]_0^x = 1 - e^{-\lambda x}.$$

As $x \to \infty$, $e^{-\lambda x} \to 0$ so $F(x) \to 1$.

EXEMPLE **CDF of a Triangular Distribution**

Let X have the symmetric triangular PDF on $[0, 2]$:

$$f(x) = \begin{cases} x, & 0 \le x \le 1, \\ 2 - x, & 1 < x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find $F(x)$.
Solution:

- $x < 0$: $F(x) = 0$.
- $0 \le x \le 1$: $F(x) = \int_0^x t\,dt = \frac{x^2}{2}$.
- $1 < x \le 2$:

$$F(x) = \int_0^1 t\,dt + \int_1^x (2 - t)\,dt = \frac{1}{2} + \left[2t - \frac{t^2}{2}\right]_1^x = \frac{1}{2} + \left(2x - \frac{x^2}{2}\right) - \left(2 - \frac{1}{2}\right) = 2x - \frac{x^2}{2} - 1.$$

- $x > 2$: $F(x) = 1$.

Thus

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{x^2}{2}, & 0 \le x \le 1, \\ 2x - \frac{x^2}{2} - 1, & 1 < x \le 2, \\ 1, & x > 2. \end{cases}$$

3.5 Using the CDF to Calculate Probabilities

The main practical use of the CDF is to compute probabilities of intervals without integrating every time.

PROPERTY **Probability via CDF**

For any continuous random variable X with CDF F, and any $a < b$,

$$P(a < X \le b) = F(b) - F(a).$$

In particular,

- $P(X \le x) = F(x)$,
- $P(X > x) = 1 - F(x)$,
- $P(a < X < b) = F(b) - F(a)$ (since $P(X = a) = P(X = b) = 0$).

WORKED EXAMPLE 6 **Using the Exponential CDF**

The lifetime X (in years) of a certain brand of solar panel installed in a village near Buea follows an exponential distribution with mean 5 years. Find:

1. The probability that a panel lasts more than 7 years.
2. The probability that it lasts between 3 and 8 years.
3. The median lifetime.

Solution: Mean $= 1/\lambda = 5 \Rightarrow \lambda = 0.2$. The CDF is $F(x) = 1 - e^{-0.2x}$ for $x \geq 0$.

1. $P(X > 7) = 1 - F(7) = e^{-0.2\times 7} = e^{-1.4} \approx 0.2466$.
2. $P(3 < X < 8) = F(8) - F(3) = (1 - e^{-1.6}) - (1 - e^{-0.6}) = e^{-0.6} - e^{-1.6} \approx 0.5488 - 0.2019 = 0.3469$.
3. The median m satisfies $F(m) = 0.5 \Rightarrow 1 - e^{-0.2m} = 0.5 \Rightarrow e^{-0.2m} = 0.5 \Rightarrow -0.2m = \ln 0.5 \Rightarrow m = \frac{\ln 2}{0.2} = 5\ln 2 \approx 3.466$ years.

3.6 Median, Quartiles and Quantiles

The CDF allows us to find the values that split the distribution into equal probability regions. These are called **quantiles**.

DEFINITION **Quantiles, Median, Quartiles**

Let X be a continuous random variable with CDF F. For any $p \in (0,1)$, the p-quantile (or 100pth percentile) is the value q_p such that

$$F(q_p) = p \quad \Longleftrightarrow \quad P(X \leq q_p) = p.$$

Special cases:

- The median is the 0.5-quantile: $F(m) = 0.5$.
- The lower quartile Q_1 is the 0.25-quantile: $F(Q_1) = 0.25$.
- The upper quartile Q_3 is the 0.75-quantile: $F(Q_3) = 0.75$.
- The interquartile range (IQR) is $Q_3 - Q_1$.

Because F is continuous and strictly increasing on the support, the quantile is unique.

METHOD **Finding Quantiles from the CDF**

1. Write down the expression for $F(x)$ on the relevant interval.

2. Set $F(x) = p$ and solve for x.
3. Verify that the solution lies within the support of X.

EXEMPLE **Median and Quartiles of the Uniform Distribution**

Let $X \sim U(a, b)$. Find the median, Q_1, Q_3 and the IQR.
Solution: $F(x) = \frac{x-a}{b-a}$ for $a \le x \le b$.

- Median m: $\frac{m-a}{b-a} = 0.5 \Rightarrow m = a + 0.5(b - a) = \frac{a+b}{2}$.
- Q_1: $\frac{Q_1-a}{b-a} = 0.25 \Rightarrow Q_1 = a + 0.25(b - a)$.
- Q_3: $\frac{Q_3-a}{b-a} = 0.75 \Rightarrow Q_3 = a + 0.75(b - a)$.
- IQR $= Q_3 - Q_1 = 0.5(b - a)$.

For the uniform distribution, the median is the midpoint, and the IQR is exactly half the range.

EXEMPLE **Median and Quartiles of the Exponential Distribution**

Let $X \sim \text{Exp}(\lambda)$. Find the median and the IQR.
Solution: $F(x) = 1 - e^{-\lambda x}$ for $x \ge 0$.

- Median m: $1 - e^{-\lambda m} = 0.5 \Rightarrow e^{-\lambda m} = 0.5 \Rightarrow m = \frac{\ln 2}{\lambda}$.
- Q_1: $1 - e^{-\lambda Q_1} = 0.25 \Rightarrow e^{-\lambda Q_1} = 0.75 \Rightarrow Q_1 = \frac{\ln(4/3)}{\lambda}$.
- Q_3: $1 - e^{-\lambda Q_3} = 0.75 \Rightarrow e^{-\lambda Q_3} = 0.25 \Rightarrow Q_3 = \frac{\ln 4}{\lambda}$.
- IQR $= Q_3 - Q_1 = \frac{\ln 4 - \ln(4/3)}{\lambda} = \frac{\ln 3}{\lambda}$.

Note that for the exponential distribution, the median $\frac{\ln 2}{\lambda}$ is less than the mean $\frac{1}{\lambda}$, reflecting the positive skew.

EXEMPLE **Median and Quartiles of a Triangular Distribution**

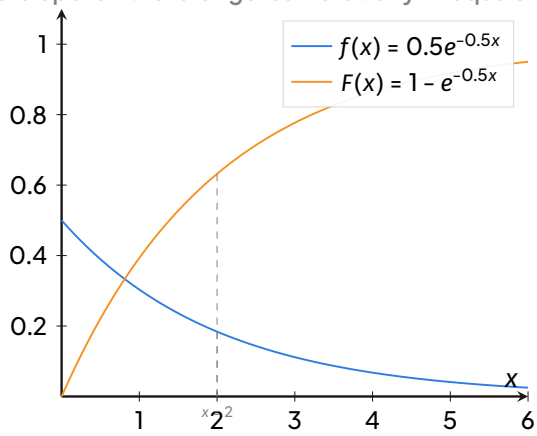
Let X have the triangular PDF on $[0, 2]$ from the earlier example: $f(x) = x$ on $[0, 1]$, $f(x) = 2 - x$ on $(1, 2]$, with CDF $F(x) = \frac{x^2}{2}$ on $[0, 1]$ and $F(x) = 2x - \frac{x^2}{2} - 1$ on $(1, 2]$. Find the median and the quartiles.
Solution: Since $F(1) = 0.5$, the median is exactly $m = 1$ (the mode, by symmetry). For Q_1: we need $F(Q_1) = 0.25$. Since $0.25 < 0.5$, Q_1 lies in $[0, 1]$. $\frac{Q_1^2}{2} = 0.25 \Rightarrow Q_1^2 = 0.5 \Rightarrow Q_1 = \frac{1}{\sqrt{2}} \approx 0.707$.

For Q_3: we need $F(Q_3) = 0.75$. Since $0.75 > 0.5$, Q_3 lies in $(1, 2]$. $2Q_3 - \frac{Q_3^2}{2} - 1 = 0.75 \Rightarrow 4Q_3 - Q_3^2 - 2 = 1.5 \Rightarrow Q_3^2 - 4Q_3 + 3.5 = 0$. $Q_3 = \frac{4 \pm \sqrt{16-14}}{2} = 2 \pm \frac{\sqrt{2}}{2}$. The root in $(1, 2]$ is $Q_3 = 2 - \frac{\sqrt{2}}{2} \approx 1.293$. IQR $= Q_3 - Q_1 = 2 - \sqrt{2} \approx 0.586$.

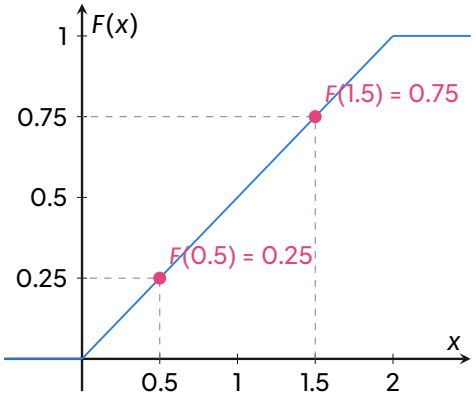
3.7 Graphical Interpretation

The CDF $F(x)$ is the area under the PDF curve to the left of x. Graphically, as x increases, the shaded area grows from 0 to 1. The slope of the CDF at any point is exactly the value of the PDF at that point: $F'(x) = f(x)$. Where the PDF is high, the CDF rises steeply; where the PDF is zero, the CDF is flat. The median is the x-coordinate where the CDF curve crosses the horizontal line $y = 0.5$; quartiles correspond to $y = 0.25$ and $y = 0.75$.

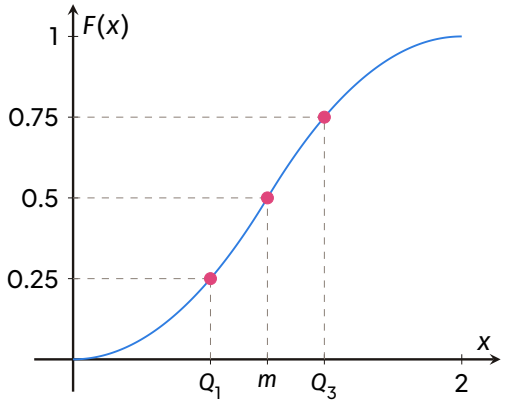
Exponential PDF $f(x) = 0.5e^{-0.5x}$ (blue) and its CDF $F(x) = 1 - e^{-0.5x}$ (orange) on the same axes. The slope of the orange curve at any x equals the height of the blue curve.



CDF of the uniform distribution on $[0, 2]$: $F(x) = \frac{x}{2}$ for $0 \le x \le 2$. The points $F(0.5) = 0.25$ and $F(1.5) = 0.75$ are marked.



CDF of the symmetric triangular distribution on $[0, 2]$. The median is at $x = 1$ where $F = 0.5$; Q_1 and Q_3 are symmetric about the median.



DID YOU KNOW? **Quantiles in Practice**

In Cameroon's meteorological services, quantiles of rainfall distributions are used to define "normal", "below normal" and "above normal" seasons. The lower quartile (25th percentile) often marks the threshold for drought warnings, while the upper quartile (75th percentile) may trigger flood preparedness. The median rainfall is a more robust measure of "typical" rainfall than the mean, especially in regions with highly skewed distributions like the Far North.

KEY POINTS **Key Points about the CDF**

- The CDF $F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt$ gives the cumulative probability up to x.
- $F(x)$ is continuous, non-decreasing, with $F(-\infty) = 0$ and $F(+\infty) = 1$.
- The PDF is the derivative of the CDF: $f(x) = F'(x)$ wherever F is differentiable.
- Probabilities of intervals are obtained by subtraction: $P(a < X \le b) = F(b) - F(a)$.
- For the uniform distribution on $[a, b]$: $F(x) = \frac{x-a}{b-a}$ on $[a, b]$.
- For the exponential distribution with rate λ: $F(x) = 1 - e^{-\lambda x}$ for $x \ge 0$.
- Quantiles are found by solving $F(q_p) = p$. Median: $p = 0.5$; Q_1: $p = 0.25$; Q_3: $p = 0.75$.

EXERCISE 3 **Practice Exercises**

- A random variable X has PDF $f(x) = \frac{3}{8}x^2$ for $0 \le x \le 2$, zero elsewhere. Find $F(x)$ and use

it to compute $P(0.5 < X < 1.5)$.

- The CDF of a continuous random variable Y is given by

$$F(y) = \begin{cases} 0, & y < 0, \\ \frac{y^2}{4}, & 0 \le y \le 2, \\ 1, & y > 2. \end{cases}$$

Find the PDF $f(y)$, the mode, and $P(Y > 1)$.

- The time (in minutes) a student spends on a Further Mathematics problem follows an exponential distribution with mean 10 minutes. Find the probability that the student spends (a) less than 5 minutes, (b) between 5 and 15 minutes, (c) more than 20 minutes. Find also the median time and the interquartile range.
- A random variable Z has a triangular PDF on $[0, 4]$ with mode at 2: $f(z) = \frac{z}{4}$ for $0 \le z \le 2$ and $f(z) = \frac{4-z}{4}$ for $2 < z \le 4$. Derive the CDF $F(z)$ and sketch both $f(z)$ and $F(z)$ on the same axes. Find the median and the quartiles of Z.
- A component's lifetime T (in hours) has CDF $F(t) = 1 - \frac{1}{(t+1)^2}$ for $t \ge 0$. Find the PDF, the mean lifetime, and the probability that the component lasts between 1 and 3 hours.

SOLUTION **Exercise 1**

$F(x) = 0$ for $x < 0$; for $0 \le x \le 2$, $F(x) = \int_0^x \frac{3}{8}t^2\,dt = \frac{x^3}{8}$; for $x > 2$, $F(x) = 1$. Then $P(0.5 < X < 1.5) = F(1.5) - F(0.5) = \frac{1.5^3}{8} - \frac{0.5^3}{8} = \frac{3.375-0.125}{8} = \frac{3.25}{8} = 0.40625$.

SOLUTION **Exercise 2**

$f(y) = F'(y) = \frac{y}{2}$ for $0 < y < 2$, zero elsewhere. Mode is at $y = 2$ (endpoint, PDF increasing). $P(Y > 1) = 1 - F(1) = 1 - \frac{1}{4} = 0.75$.

SOLUTION **Exercise 3**

$\lambda = 1/10 = 0.1$. $F(t) = 1 - e^{-0.1t}$. **(a)** $P(T < 5) = F(5) = 1 - e^{-0.5} \approx 0.3935$. **(b)** $P(5 < T < 15) = F(15) - F(5) = e^{-0.5} - e^{-1.5} \approx 0.6065 - 0.2231 = 0.3834$. **(c)** $P(T > 20) = 1 - F(20) = e^{-2} \approx 0.1353$. Median $m = \frac{\ln 2}{0.1} = 10 \ln 2 \approx 6.93$ minutes. $Q_1 = \frac{\ln(4/3)}{0.1} \approx 2.88$ minutes, $Q_3 = \frac{\ln 4}{0.1} \approx 13.86$ minutes. IQR $= Q_3 - Q_1 = \frac{\ln 3}{0.1} \approx 10.99$ minutes.

SOLUTION **Exercise 4**

For $0 \le z \le 2$: $F(z) = \int_0^z \frac{t}{4}\,dt = \frac{z^2}{8}$. For $2 < z \le 4$: $F(z) = F(2) + \int_2^z \frac{4-t}{4}\,dt = \frac{1}{2} + \left[t - \frac{t^2}{8}\right]_2^z = \frac{1}{2} + \left(z - \frac{z^2}{8}\right) - \left(2 - \frac{1}{2}\right) = z - \frac{z^2}{8} - 1$. Check: $F(4) = 4 - 2 - 1 = 1$. Median: $F(2) = 0.5 \Rightarrow m = 2$. Q_1: $\frac{Q_1^2}{8} = 0.25 \Rightarrow Q_1^2 = 2 \Rightarrow Q_1 = \sqrt{2} \approx 1.414$. Q_3: $Q_3 - \frac{Q_3^2}{8} - 1 = 0.75 \Rightarrow Q_3^2 - 8Q_3 + 14 = 0 \Rightarrow Q_3 = 4 - \sqrt{2} \approx 2.586$. IQR $= 4 - 2\sqrt{2} \approx 1.172$. The graph of f is a triangle peaking at $(2, 0.5)$; the graph of F is a smooth S-shaped curve (quadratic pieces) from $(0, 0)$ to $(4, 1)$ with inflection at the mode $z = 2$.

SOLUTION **Exercise 5**

$f(t) = F'(t) = \frac{2}{(t+1)^3}$ for $t \ge 0$. Mean: $E(T) = \int_0^\infty t \cdot \frac{2}{(t+1)^3}\,dt$. Let $u = t + 1$, then $t = u - 1$, $dt = du$: $E(T) = \int_1^\infty \frac{2(u-1)}{u^3}\,du = 2\int_1^\infty (u^{-2} - u^{-3})\,du = 2\left[-u^{-1} + \frac{1}{2}u^{-2}\right]_1^\infty = 2(0 - (-1 + 0.5)) = 1$ hour. $P(1 < T < 3) = F(3) - F(1) = \left(1 - \frac{1}{16}\right) - \left(1 - \frac{1}{4}\right) = \frac{1}{4} - \frac{1}{16} = \frac{3}{16} = 0.1875$.

4 MEDIAN, QUARTILES AND QUANTILES – APPLICATIONS

In the previous section, we introduced the cumulative distribution function $F(x) = \mathbb{P}(X \le x)$, a central tool for calculating interval probabilities. We will now exploit this function to locate characteristic values of the distribution: the median, quartiles, and more generally, quantiles. These position indicators are essential for summarising a continuous distribution, comparing samples, or making decisions (warranty thresholds, alert thresholds, etc.).

4.1 Median, Quartiles and Quantiles of Order p: Definitions and Interpretation

Recall that for a continuous random variable X with cumulative distribution function F, strictly increasing on its support, the equation $F(x) = p$ has a unique solution for every $p \in (0, 1)$.

DEFINITION **Median, Quartiles and Quantiles**

Let X be a continuous random variable with cumulative distribution function F.

- The median m (or Q_2) is the value such that $\mathbb{P}(X \le m) = \mathbb{P}(X \ge m) = 0.5$, i.e. $F(m) = 0.5$.
- The first quartile Q_1 satisfies $F(Q_1) = 0.25$; the third quartile Q_3 satisfies $F(Q_3) = 0.75$.
- More generally, for any $p \in (0, 1)$, the quantile of order p (or p-quantile), denoted q_p or $Q(p)$, is the unique solution of $F(q_p) = p$.

KEY POINTS **Interpretation**

- The median splits the area under the density curve f into two equal parts.
- Q_1 leaves 25% of the probability to its left, Q_3 leaves 75%.
- The interquartile range $IQR = Q_3 - Q_1$ measures the central dispersion (robust to extreme values).

WARNING **Common Pitfalls**

- Do not confuse the median (quantile of order 0.5) with the expectation (mean). They coincide only for symmetric distributions.
- Solving $F(x) = p$ may yield several algebraic roots; keep only the one belonging to the support of X.
- Verify that F is indeed continuous and strictly increasing on the interval considered; otherwise, the quantile may not be unique (case of piecewise constant distributions).

4.2 Useful Properties and the Case of Symmetric Distributions

PROPERTY **Symmetric Distribution**

If the density f is symmetric about the axis $x = a$ (i.e. $f(a + x) = f(a - x)$ for all x), then:

$$\mathbb{E}(X) = a, \quad \text{mode} = a, \quad \text{median} = a.$$

In particular, for a uniform distribution on $[a, b]$, a normal distribution, or a symmetric triangular distribution, mean = median = mode = centre of symmetry.

PROPERTY **Quantiles under Affine Transformation**

If $Y = aX + b$ with $a > 0$, then for any $p \in (0, 1)$:

$$q_p(Y) = a\, q_p(X) + b.$$

If $a < 0$, the order is reversed: $q_p(Y) = a\, q_{1-p}(X) + b$.

4.3 Resolution Method: Finding the Quantile of Order p

METHOD **Calculating a Quantile q_p**

1. Write the cumulative distribution function $F(x)$ for all x in the support (by integrating f).
2. Solve the equation $F(x) = p$ for x.
3. Verify that the solution found belongs to the support of X.
4. (Optional) Graphical check: plot $y = F(x)$ and the horizontal line $y = p$; the abscissa of the intersection is q_p.

4.4 Explicit Calculations for Classical Distributions

Uniform Distribution on $[a, b]$. $f(x) = \frac{1}{b-a}\mathbf{1}_{[a,b]}(x)$, $F(x) = \frac{x-a}{b-a}$ for $x \in [a, b]$.

$$q_p = a + p(b - a), \quad m = \frac{a + b}{2}, \quad Q_1 = a + \frac{1}{4}(b - a), \quad Q_3 = a + \frac{3}{4}(b - a).$$

Exponential Distribution with Parameter $\lambda > 0$. $f(x) = \lambda e^{-\lambda x}\mathbf{1}_{[0,+\infty[}(x)$, $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$.

$$1 - e^{-\lambda q_p} = p \implies q_p = -\frac{\ln(1 - p)}{\lambda}.$$

Median: $m = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}$. $Q_1 = -\frac{\ln 0.75}{\lambda} \approx \frac{0.288}{\lambda}$, $Q_3 = -\frac{\ln 0.25}{\lambda} \approx \frac{1.386}{\lambda}$. Remark: $m < \mathbb{E}(X) = 1/\lambda$ (right-skewed asymmetry).

Symmetric Triangular Distribution on $[0, 2]$ with Mode at 1. $f(x) = \begin{cases} x & 0 \leq x \leq 1 \\ 2 - x & 1 \leq x \leq 2 \end{cases}$. $F(x) = \begin{cases} \frac{x^2}{2} & 0 \leq x \leq \\ 1 - \frac{(2-x)^2}{2} & 1 \leq x \leq 2 \end{cases}$

By symmetry: $m = 1 = \mathbb{E}(X)$. Q_1: solve $\frac{x^2}{2} = 0.25 \Rightarrow x = \frac{1}{\sqrt{2}} \approx 0.707$. Q_3: by symmetry $Q_3 = 2 - Q_1 = 2 - \frac{1}{\sqrt{2}} \approx 1.293$.

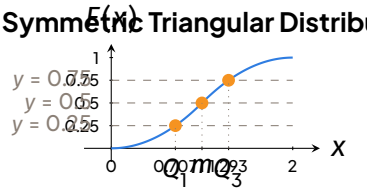
EXEMPLE **Quantiles of an Asymmetric Triangular Distribution**

Let X have density $f(x) = 2x\,\mathbf{1}_{[0,1]}(x)$ (triangular on $[0, 1]$ with mode at 1). $F(x) = x^2$ for $x \in [0, 1]$. Median: $m^2 = 0.5 \Rightarrow m = \sqrt{0.5} = \frac{\sqrt{2}}{2} \approx 0.707$. Q_1: $x^2 = 0.25 \Rightarrow x = 0.5$. Q_3: $x^2 = 0.75 \Rightarrow x = \sqrt{0.75} = \frac{\sqrt{3}}{2} \approx 0.866$. Expectation: $\mathbb{E}(X) = \int_0^1 2x^2\,dx = \frac{2}{3} \approx 0.667$. We observe $Q_1 < \mathbb{E}(X) < m < Q_3$, typical of a right-skewed asymmetry.

4.5 Graphical Interpretation: CDF and Horizontal Lines

The figure below illustrates the graphical search for quantiles for the symmetric triangular distribution on $[0, 2]$. The intersections of the curve $y = F(x)$ with the horizontal lines $y = 0.25$,

$y = 0.5$, $y = 0.75$ give Q_1, m, Q_3.



Graphical search for the median and quartiles by intersection of $F(x)$ with horizontal lines.

4.6 Complete Application Problem: Modelling Taxi Waiting Time

Consider a concrete situation: in Douala, the waiting time (in minutes) for a taxi at a given junction is modelled by a continuous random variable X with density:

$$f(x) = \begin{cases} k\,x\,(10 - x) & \text{if } 0 \le x \le 10, \\ 0 & \text{otherwise.} \end{cases}$$

WORKED EXAMPLE 7 **Complete Analysis of Waiting Time**

1. Determine the constant k (normalisation).

$$\int_0^{10} k\,x(10 - x)\,dx = k\int_0^{10} (10x - x^2)\,dx = k\left[5x^2 - \frac{x^3}{3}\right]_0^{10} = k\left(500 - \frac{1000}{3}\right) = k \cdot \frac{500}{3} = 1$$

Hence $k = \frac{3}{500} = 0.006$.

2. Cumulative Distribution Function $F(x)$ for $0 \le x \le 10$.

$$F(x) = \int_0^x \frac{3}{500}\,t(10 - t)\,dt = \frac{3}{500}\left[5t^2 - \frac{t^3}{3}\right]_0^x = \frac{3}{500}\left(5x^2 - \frac{x^3}{3}\right) = \frac{15x^2 - x^3}{500}.$$

Check: $F(10) = \frac{1500-1000}{500} = 1$.

3. Expectation and Variance.

$$\mathbb{E}(X) = \int_0^{10} x f(x)\,dx = \frac{3}{500}\int_0^{10} (10x^2 - x^3)\,dx = \frac{3}{500}\left[\frac{10x^3}{3} - \frac{x^4}{4}\right]_0^{10} = \frac{3}{500}\left(\frac{10000}{3} - 2500\right) = \frac{3}{500} \cdot \frac{2500}{3} = 5$$

$$\mathbb{E}(X^2) = \frac{3}{500}\int_0^{10} (10x^3 - x^4)\,dx = \frac{3}{500}\left[\frac{10x^4}{4} - \frac{x^5}{5}\right]_0^{10} = \frac{3}{500}(25000 - 20000) = 30.$$

$$\mathrm{Var}(X) = 30 - 5^2 = 5\ \text{min}^2, \quad \sigma = \sqrt{5} \approx 2.236\ \text{min}.$$

4. Mode. $f(x) = \frac{3}{500}(10x - x^2)$ is a concave quadratic, maximum at $f'(x) = 0 \Rightarrow 10 - 2x = 0 \Rightarrow x = 5$. Mode = 5 min. The distribution is symmetric (parabolic density centred at 5), so mean = median = mode = 5.

5. Median and Quartiles. By symmetry: $m = 5$. Q_1: solve $F(x) = 0.25 \Rightarrow 15x^2 - x^3 = 125 \Rightarrow x^3 - 15x^2 + 125 = 0$. Seek a root in $[0, 5]$. Trial: $x = 2.5 \Rightarrow 15.625 - 93.75 + 125 = 46.875$;

$x = 3 \Rightarrow 27 - 135 + 125 = 17$; $x = 3.5 \Rightarrow 42.875 - 183.75 + 125 = -15.875$. The root lies between 3 and 3.5. Using a calculator (or numerical method): $Q_1 \approx 3.23$ min. By symmetry: $Q_3 = 10 - Q_1 \approx 6.77$ min. Interquartile range $IQR = Q_3 - Q_1 \approx 3.54$ min.
6. Probability of Waiting More Than 10 Minutes. The support is $[0, 10]$, so $\mathbb{P}(X > 10) = 0$. (Common trap question: check the support!)
7. Probability of Waiting Between 3 and 7 Minutes. $\mathbb{P}(3 \le X \le 7) = F(7) - F(3)$. $F(7) = \frac{15 \cdot 49 - 343}{500} = \frac{735 - 343}{500} = \frac{392}{500} = 0.784$. $F(3) = \frac{15 \cdot 9 - 27}{500} = \frac{135 - 27}{500} = \frac{108}{500} = 0.216$. Probability = $0.784 - 0.216 = 0.568$ (i.e. 56.8%).

SOLUTION

Pedagogical Analysis:

- Normalisation (k) is the mandatory first step.
- The symmetry of f (parabola centred at 5) greatly simplifies calculations: $\mathbb{E}(X) = m =$ mode $= 5$.
- For Q_1, the cubic equation does not factorise simply; using a calculator (numerical solver) is expected at GCE. Give the approximate value to 2 or 3 decimal places.
- Always verify that the quantiles found belong to the support $[0, 10]$.
- The question $\mathbb{P}(X > 10)$ tests understanding of the support.

4.7 GCE-Type Exercise Resolution Strategies

METHOD **Checklist for a Continuous Random Variable Exercise**

1. Identify the distribution: uniform, exponential, triangular, or explicitly given density (piecewise or not).
2. Check normalisation: $\int_{-\infty}^{+\infty} f(x)\,dx = 1$. If an unknown constant (k, λ, etc.) is present, determine it first.
3. Establish the CDF $F(x)$ by integration, distinguishing intervals if f is piecewise. Check continuity of F at junctions.
4. Calculate requested characteristics:
   - Expectation: $\int x f(x)\,dx$.
   - Variance: $\mathbb{E}(X^2) - [\mathbb{E}(X)]^2$.
   - Mode: maximum of $f(x)$ (derivative or graphical observation).
   - Median/quartiles/quantiles: solve $F(x) = p$.

5. Calculate requested probabilities: $\mathbb{P}(a < X < b) = F(b) - F(a)$.
6. Check consistency: support respected, probabilities in $[0, 1]$, median between Q_1 and Q_3, positive variance.

WARNING **Classic Errors to Avoid at GCE**

- Forgetting to multiply by the normalisation constant k in expectation/variance integrals.
- Confusing $f(x)$ (density) and $F(x)$ (CDF): the median is found with F, the mode with f.
- Solving $f(x) = p$ instead of $F(x) = p$ for a quantile.
- Not verifying that the solution of $F(x) = p$ belongs to the support (e.g., negative root for an exponential distribution).
- For a piecewise density, integrate each piece separately and sum; do not integrate the global expression over $\mathbb{R}$.
- Writing $\mathbb{P}(X = x)$ for a continuous variable (it is always 0).

4.8 Practice Exercises

EXERCISE 4 **Exponential Distribution and Warranty Thresholds**

The lifetime X (in years) of an electronic component follows an exponential distribution with parameter $\lambda = 0.2$.

1. Determine the expectation, variance, and median of X.
2. Calculate Q_1 and Q_3. Interpret the interquartile range.
3. The manufacturer wants to offer a warranty such that 90% of components are still functional at the end of the period. What warranty duration should be proposed?
4. What is the probability that a component lasts more than 10 years given that it has already lasted 5 years? (Memoryless property).

EXERCISE 5 **Piecewise Density – Flow Rate Model**

The flow rate X (in L min^{-1}) of a water pump follows the density:

$$f(x) = \begin{cases} c\,x & 0 \le x \le 2, \\ c\,(4 - x) & 2 < x \le 4, \\ 0 & \text{otherwise.} \end{cases}$$

1. Find c and sketch $f(x)$.
2. Calculate $\mathbb{E}(X)$, $\mathrm{Var}(X)$, and the mode.
3. Determine the median and quartiles.
4. Calculate $\mathbb{P}(X > 3)$.

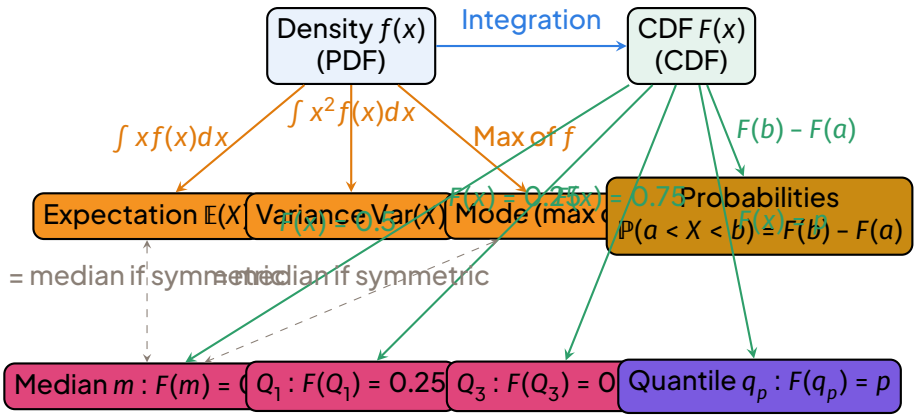
EXERCISE 6 **Quantiles and Affine Transformation**

Let $X \sim U[0, 1]$ (uniform on $[0, 1]$). Define $Y = 5X + 2$.

1. Give the density of Y and its support.
2. Express the quantiles of Y in terms of those of X.
3. Calculate directly the median, Q_1, and Q_3 of Y without using X.

4.9 Visual Summary: From PDF to Quantiles

The diagram below summarises the links between all concepts covered in this chapter (sections 1 to 4). It shows how the density f generates the CDF F, and how we extract numerical characteristics (expectation, variance, mode) and position indicators (median, quartiles, quantiles).



Mind map linking density, CDF, moments, and quantiles.

DID YOU KNOW? **Applications in Cameroon and Beyond**

- Public Health: Quartiles of waiting time distributions in emergency departments (Laquintinie Hospital, Central Hospital) allow setting quality targets (e.g., "75% of patients seen in under 30 min" $\Leftrightarrow Q_3 \leq 30$).
- Agriculture: The distribution of yields per hectare (maize, cocoa) is often skewed; the median is more representative than the mean for the average farmer.
- Telecoms: Operators (MTN, Orange) monitor latency quantiles (e.g., $Q_{0.95}$) to guarantee QoS (Quality of Service).
- Finance: Value-at-Risk (VaR) at 99% is exactly the quantile of order 0.01 of the loss distribution (left tail).

KEY POINTS **Key Points for the GCE Exam**

- Median m: solve $F(m) = 0.5$.
- Quartiles Q_1, Q_3: solve $F(x) = 0.25$ and $F(x) = 0.75$.
- Quantile of order p: solve $F(q_p) = p$.
- Always verify that the solution belongs to the support.
- For symmetric distributions: mean = median = mode = centre of symmetry.
- Uniform distribution: $q_p = a + p(b - a)$.
- Exponential distribution: $q_p = -\frac{\ln(1-p)}{\lambda}$.
- Triangular distribution: integrate piecewise, solve the polynomial equation (often 2nd or 3rd degree).
- The interquartile range $IQR = Q_3 - Q_1$ is a robust measure of dispersion.

5 INTEGRATION ACTIVITY

6 INTEGRATION ACTIVITY: WAITING FOR A TAXI IN YAOUNDÉ

6.1 Aims of the Activity

This integration activity closes the chapter on **continuous random variables**. Working in groups, you will mobilise, in a single realistic situation, all the notions studied:

- verify and normalise a **probability density function** (PDF) by computing the constant k from the condition $\int_{-\infty}^{+\infty} f(x)\,dx = 1$;
- derive the **cumulative distribution function** (CDF) F from f by integration, and use it to compute probabilities of intervals;
- compute the **expectation** $E(X)$ and the **variance** $\mathrm{Var}(X)$ of a continuous random variable, using integration by parts;
- determine the **mode** (maximum of f), the **median** and the **quartiles** (solutions of $F(x) = q$);
- interpret every numerical result concretely in terms of waiting time, and criticise a mathematical model by comparing it with real survey data.

Beyond the calculations, the activity trains you in the complete modelling cycle: situation → model → computation → interpretation → validation. This is exactly the spirit of the GCE Advanced Level problem-solving questions.

6.2 Situation

Waiting for a taxi in Yaoundé.

The Students' Association of a large secondary school in Yaoundé carries out a study on the daily journey of its members. Every morning, hundreds of students wait at the taxi rank near the Mokolo market to board a shared taxi (clando or yellow taxi) towards their school. The association suspects that the waiting time is highly variable: some passengers board almost immediately, others wait a very long time, and this irregularity causes lateness.

A statistics teacher suggests modelling the situation. Let X be the continuous random variable equal to the **waiting time, in minutes**, of a passenger chosen at random at the rank. Based on the shape of the histogram drawn from a preliminary observation (a curve starting at 0, rising to a peak, then decreasing with a long tail), the teacher proposes the family of density functions:

$$f(x) = \begin{cases} k\,x\,e^{-x/5} & \text{if } x \geq 0, \\ 0 & \text{if } x < 0, \end{cases}$$

where k is a positive real constant to be determined.

The association also conducted a small survey of 200 passengers during one week. The observed data are summarised below:

Waiting time (min)	[0; 5[	[5; 10[	[10; 15[	[15; 20[	[20; +∞[
Number of passengers	54	62	44	24	16

The mean waiting time observed in the survey is about 9.9 minutes, and 40 passengers out of 200 waited more than 12 minutes.

Your mission: as members of the mathematics club, use the model f to describe the waiting time completely (typical value, average value, spread, probabilities), then discuss whether the model fits the survey data. You will present your findings orally to the association.

WARNING **Data reminder**

All the data needed are inside the statement above. The integrals you will need are standard at Upper Sixth level: integration by parts for $\int xe^{-x/5}dx$ and $\int x^2e^{-x/5}dx$, and the limit $\lim_{x\to+\infty} x^n e^{-x/5} = 0$ for any fixed n (exponential dominates polynomial).

6.3 Tasks

Work in groups of three or four. Answer the questions in order; each one uses a different tool of the chapter.

1. **Normalising the density.** Show that f is a probability density function if and only if $k\int_0^{+\infty} xe^{-x/5}\,dx = 1$. Using integration by parts, compute this integral and deduce the value of k.

2. **The cumulative distribution function.** Show that, for $x \geq 0$,

$$F(x) = 1 - \left(1 + \frac{x}{5}\right)e^{-x/5}.$$

   State $F(x)$ for $x < 0$. Verify that $\lim_{x\to+\infty} F(x) = 1$.

3. **Graphs.** On the same page, sketch the curves of f and F (two separate axes systems, same scale in x). Indicate on the graph of f the region whose area equals $P(X > 12)$.

4. **Expectation and variance.** Compute $E(X)$ and $E(X^2)$. Deduce $\mathrm{Var}(X)$ and the standard deviation $\sigma(X)$, rounded to two decimal places. Interpret $E(X)$ in the context.

5. **Mode.** Determine the mode of X (the value of x where f is maximal). Why is the mode smaller than the mean here?

6. **Median and quartiles.** Using the equation $F(x) = q$ with a calculator (trial or solver), determine the median m, the first quartile Q_1 and the third quartile Q_3, rounded to one decimal place. Deduce the interquartile range.

7. **A probability of interest.** Compute $P(X > 12)$, the probability that a passenger waits more than 12 minutes. Give the result to three decimal places.

8. **Model versus reality.** Compare with the survey: (a) the model mean with the observed mean 9.9 min; (b) $P(X > 12)$ with the observed proportion 40/200; (c) the model's probabilities of the five classes of the table with the observed frequencies. Conclude: is the model acceptable? Suggest one improvement.

METHOD **Oral Restitution**

Each group prepares a five-minute presentation: one member presents the model and the graphs, one presents the numerical characteristics (mean, variance, mode, median, quartiles), one presents the comparison with the survey and the conclusion. Every numerical result must be interpreted in a sentence about waiting times, not merely stated.

6.4 Model Answer

1. Determination of k.

f is continuous on $\mathbb{R}$ and non-negative since $k > 0$ and $xe^{-x/5} \geq 0$ for $x \geq 0$. It is a density if and only if its total integral equals 1. Integrating by parts with $u = x$, $dv = e^{-x/5}dx$:

$$\int_0^{+\infty} xe^{-x/5}\,dx = \left[-5xe^{-x/5}\right]_0^{+\infty} + 5\int_0^{+\infty} e^{-x/5}\,dx = 0 + 5\left[-5e^{-x/5}\right]_0^{+\infty} = 25.$$

Hence $25k = 1$, so $\boxed{k = \frac{1}{25} = 0.04}$ and $f(x) = \frac{x}{25}e^{-x/5}$ for $x \geq 0$.

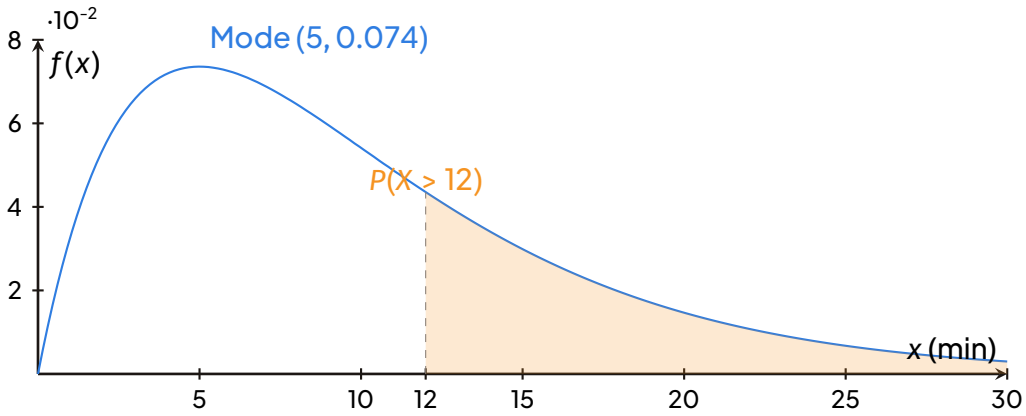
2. The CDF.

For $x < 0$, $F(x) = 0$. For $x \geq 0$, integrating by parts again:

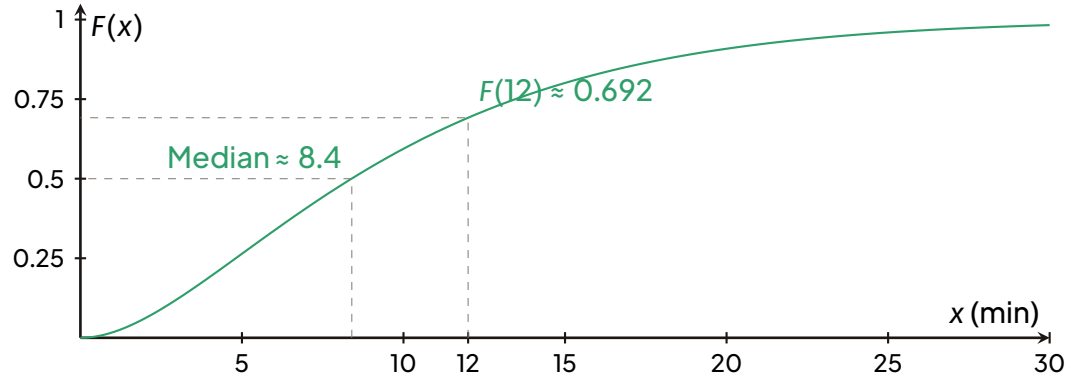
$$F(x) = \frac{1}{25}\int_0^x te^{-t/5}\,dt = \frac{1}{25}\left(\left[-5te^{-t/5}\right]_0^x + 25\left(1 - e^{-x/5}\right)\right) = 1 - \left(1 + \frac{x}{5}\right)e^{-x/5}.$$

Check: $F(0) = 1 - 1 = 0$ and, since $e^{-x/5}$ dominates, $\lim_{x\to+\infty} F(x) = 1$. F is continuous and increasing: it is a valid CDF.

3. Graphs.



Probability density function $f(x) = \frac{x}{25}e^{-x/5}$. The shaded area represents $P(X > 12)$.



Cumulative distribution function $F(x) = 1 - (1 + \frac{x}{5})e^{-x/5}$.

4. Expectation and variance.

$$E(X) = \frac{1}{25}\int_0^{+\infty} x^2 e^{-x/5}\,dx.$$

Two successive integrations by parts (or the standard result $\int_0^{+\infty} x^2 e^{-x/a}dx = 2a^3$ with $a = 5$) give $\int_0^{+\infty} x^2 e^{-x/5}dx = 250$, hence

$$E(X) = \frac{250}{25} = \boxed{10\text{ min}}.$$

Similarly, $\int_0^{+\infty} x^3 e^{-x/5}dx = 6 \times 5^4 = 3750$, so

$$E(X^2) = \frac{3750}{25} = 150, \qquad \mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 150 - 100 = \boxed{50\text{ min}^2},$$

and $\sigma(X) = \sqrt{50} \approx \boxed{7.07\text{ min}}$. On average a passenger waits 10 minutes, with a typical fluctuation of about 7 minutes: the waiting time is indeed very irregular, confirming the association's concern.

5. Mode.

For $x > 0$, $f'(x) = \frac{1}{25}e^{-x/5}\left(1 - \frac{x}{5}\right)$, which vanishes at $x = 5$ and changes sign from + to −. The **mode** is $\boxed{5\text{ min}}$. The mode is smaller than the mean because the distribution is right-skewed: a minority of very long waits pulls the mean upwards without affecting the peak.

6. Median and quartiles.

We solve $F(x) = q$, i.e. $\left(1 + \frac{x}{5}\right)e^{-x/5} = 1 - q$, numerically:

- $F(Q_1) = 0.25$ gives $Q_1 \approx 4.8$ min;
- $F(m) = 0.5$ gives $m \approx 8.4$ min;
- $F(Q_3) = 0.75$ gives $Q_3 \approx 13.5$ min.

Interquartile range: $Q_3 - Q_1 \approx 8.7$ min. Half of the passengers wait less than about 8.4 minutes, and half of all waits lie between about 4.8 and 13.5 minutes. Note the ordering mode (5) < median (8.4) < mean (10), characteristic of a right-skewed distribution.

7. Probability of waiting more than 12 minutes.

$$P(X > 12) = 1 - F(12) = \left(1 + \frac{12}{5}\right) e^{-12/5} = 3.4\, e^{-2.4} \approx 3.4 \times 0.090718 \approx \boxed{0.308}.$$

According to the model, about 30.8% of passengers wait more than 12 minutes.

8. Model versus reality.

- **Mean:** model 10.0 min versus observed 9.9 min — excellent agreement.
- $P(X > 12)$: model 0.308 versus observed $40/200 = 0.200$ — the model overestimates long waits.
- **Classes:** the model gives

$$P(0 \le X < 5) = F(5) = 1 - 2e^{-1} \approx 0.264 \quad \text{(observed } 54/200 = 0.270\text{)},$$
$$P(5 \le X < 10) = F(10) - F(5) = 2e^{-1} - 3e^{-2} \approx 0.330 \quad \text{(observed } 0.310\text{)},$$
$$P(10 \le X < 15) = F(15) - F(10) = 3e^{-2} - 4e^{-3} \approx 0.207 \quad \text{(observed } 0.220\text{)},$$
$$P(15 \le X < 20) = F(20) - F(15) = 4e^{-3} - 5e^{-4} \approx 0.108 \quad \text{(observed } 0.120\text{)},$$
$$P(X \ge 20) = 1 - F(20) = 5e^{-4} \approx 0.092 \quad \text{(observed } 0.080\text{)}.$$

The first class matches very well; the model puts slightly more weight on the $[5, 10[$ class and slightly less on the $[10, 15[$ and $[15, 20[$ classes, while the tail $[20, +\infty[$ is also slightly overestimated. **Conclusion:** the model $f(x) = \frac{x}{25} e^{-x/5}$ (a Gamma-type density with shape 2, scale 5) is acceptable for planning purposes — for instance, the school can tell students that leaving a margin of about 15 minutes covers roughly $F(15) \approx 80.1\%$ of cases. A possible improvement is to fit a general Gamma density $f(x) = kx^{\alpha-1}e^{-x/\beta}$ with two parameters, or to collect a larger sample over several weeks and different times of day.

KEY POINTS **What this activity consolidates**

- A density must satisfy $f \ge 0$ and total area 1 — this fixes any unknown constant.
- The CDF is obtained by integration: $F(x) = \int_{-\infty}^{x} f(t)\,dt$, and $P(X > a) = 1 - F(a)$.
- $E(X) = \int x f(x)\,dx$, $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$: integration by parts is the key technique.
- Mode = maximum of f; median and quartiles = solutions of $F(x) = q$, often found numerically.
- A model is only useful once confronted with real data: agreement on the mean and the classes, and honest discussion of discrepancies.

7 METHODS AND WORKED EXAMPLES

7.1 Skill 1 — Verifying that a function is a probability density function

METHOD **Testing whether f is a PDF**

To show that a function f is a **probability density function** on an interval $[a, b]$ (or on $\mathbb{R}$), proceed as follows.

1. Check non-negativity. Show that $f(x) \geq 0$ for every x in the interval (and $f(x) = 0$ outside it).
2. Check the total area. Compute $\int_a^b f(x)\,dx$ and show that it equals 1.
3. If the density contains an unknown constant k, use the condition $\int_a^b f(x)\,dx = 1$ to find k first, then check positivity.

Reflexes: probabilities are areas under the curve; for a continuous variable $P(X = c) = 0$ for any single value c, so $P(X \leq c) = P(X < c)$.

WORKED EXAMPLE 8 **Finding the constant in a density**

The continuous random variable X has probability density function

$$f(x) = \begin{cases} kx(4 - x^2), & 0 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

1. Find the value of the constant k.
2. Find $P(0.5 \leq X \leq 1.5)$.
3. Find $P(X > 1)$.

SOLUTION

1. Value of k. Since f is a density, $\int_0^2 kx(4 - x^2)\,dx = 1$.

$$\int_0^2 k(4x - x^3)\,dx = k\left[2x^2 - \frac{x^4}{4}\right]_0^2 = k\,(8 - 4) = 4k.$$

Hence $4k = 1$, so $\boxed{k = \frac{1}{4}}$. On $[0, 2]$, $x \geq 0$ and $4 - x^2 \geq 0$, so $f(x) \geq 0$: f is a genuine density.

2. $P(0.5 \le X \le 1.5)$. **Probabilities are integrals of f:**

$$P(0.5 \le X \le 1.5) = \frac{1}{4}\int_{0.5}^{1.5}(4x - x^3)\,dx = \frac{1}{4}\left[2x^2 - \frac{x^4}{4}\right]_{0.5}^{1.5}$$
$$= \frac{1}{4}\left[\left(2(2.25) - \frac{5.0625}{4}\right) - \left(2(0.25) - \frac{0.0625}{4}\right)\right]$$
$$= \frac{1}{4}\left[(4.5 - 1.265625) - (0.5 - 0.015625)\right] = \frac{1}{4}(2.75) = 0.6875.$$

3. $P(X > 1)$.

$$P(X > 1) = \frac{1}{4}\int_{1}^{2}(4x - x^3)\,dx = \frac{1}{4}\left[2x^2 - \frac{x^4}{4}\right]_{1}^{2} = \frac{1}{4}\left[4 - \left(2 - \tfrac{1}{4}\right)\right] = \frac{1}{4}\cdot\frac{9}{4} = \frac{9}{16} = 0.5625.$$

7.2 Skill 2 — Computing probabilities from a PDF

METHOD **Probabilities as areas**

Given a density f of a continuous random variable X:

1. Identify the interval on which f is non-zero; split the integral if f is defined piecewise.
2. Use $P(a \le X \le b) = \int_a^b f(x)\,dx$.
3. Use complementation when easier: $P(X > a) = 1 - P(X \le a)$.
4. Remember $P(X = a) = 0$: strict or non-strict inequalities give the same answer.

WORKED EXAMPLE 9 **Piecewise density**

The lifetime X (in years) of a phone battery sold in Douala has density

$$f(x) = \begin{cases} \frac{3}{64}x^2, & 0 \le x < 2, \\ \frac{3}{64}(8x - x^2 - 12), & 2 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

Given that $\int_2^4 \frac{3}{64}(8x - x^2 - 12)\,dx = \frac{1}{2}$, find $P(X > 3)$ and $P(X > 3 \mid X > 2)$.

SOLUTION

Step 1: $P(X > 3)$. Only the second piece is needed on $[3, 4]$:

$$P(X > 3) = \frac{3}{64}\int_3^4 (8x - x^2 - 12)\,dx = \frac{3}{64}\left[4x^2 - \frac{x^3}{3} - 12x\right]_3^4$$
$$= \frac{3}{64}\left[\left(64 - \frac{64}{3} - 48\right) - (36 - 9 - 36)\right]$$
$$= \frac{3}{64}\left[\left(16 - \frac{64}{3}\right) - (-9)\right] = \frac{3}{64}\left(-\frac{16}{3} + 9\right) = \frac{3}{64}\left(\frac{11}{3}\right) = \frac{11}{64}.$$

Step 2: conditional probability. First,

$$P(X > 2) = \int_2^4 f(x)\,dx = \frac{1}{2} \quad \text{(given)}.$$

Since $\{X > 3\} \subset \{X > 2\}$,

$$P(X > 3 \mid X > 2) = \frac{P(X > 3)}{P(X > 2)} = \frac{11/64}{1/2} = \frac{11}{32} = 0.34375.$$

7.3 Skill 3 — Expectation and variance of a continuous random variable

METHOD **Computing $E(X)$, $\mathrm{Var}(X)$ and functions of X**

1. Mean: $E(X) = \int_a^b x\,f(x)\,dx$ over the support of f.
2. Second moment: $E(X^2) = \int_a^b x^2 f(x)\,dx$.
3. Variance: $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$. Never forget to subtract the square of the mean.
4. Linear rules: $E(aX + b) = aE(X) + b$ and $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$ (the shift b does not affect variance).
5. For a function g: $E[g(X)] = \int_a^b g(x)f(x)\,dx$.

Reflex: keep fractions as long as possible; convert to decimals only at the end.

WORKED EXAMPLE 10 **Mean and variance from a density**

The continuous random variable X has density

$$f(x) = \begin{cases} \frac{3}{4}x(2-x), & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

1. Show that $E(X) = 1$.
2. Find $\mathrm{Var}(X)$.
3. Deduce $E(3X - 2)$ and $\mathrm{Var}(3X - 2)$.

SOLUTION

1. Mean.

$$E(X) = \int_0^2 x \cdot \frac{3}{4}x(2-x)\,dx = \frac{3}{4}\int_0^2 (2x^2 - x^3)\,dx$$
$$= \frac{3}{4}\left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_0^2 = \frac{3}{4}\left(\frac{16}{3} - 4\right) = \frac{3}{4}\cdot\frac{4}{3} = 1.$$

2. Variance. First the second moment:

$$E(X^2) = \frac{3}{4}\int_0^2 x^2(2x - x^2)\,dx = \frac{3}{4}\int_0^2 (2x^3 - x^4)\,dx$$
$$= \frac{3}{4}\left[\frac{x^4}{2} - \frac{x^5}{5}\right]_0^2 = \frac{3}{4}\left(8 - \frac{32}{5}\right) = \frac{3}{4}\cdot\frac{8}{5} = \frac{6}{5}.$$

Hence

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{6}{5} - 1^2 = \frac{1}{5} = 0.2.$$

3. Linear transformation.

$$E(3X - 2) = 3E(X) - 2 = 3 - 2 = 1, \qquad \mathrm{Var}(3X - 2) = 9\,\mathrm{Var}(X) = \frac{9}{5} = 1.8.$$

7.4 Skill 4 — Finding the mode

METHOD **Locating the mode**

The **mode** of a continuous random variable is the value x where the density f attains its maximum.

1. Differentiate f on its support and solve $f'(x) = 0$.
2. Justify that the stationary point is a maximum (sign of f', or $f''(x) < 0$, or comparison with endpoint values).

3. State the mode clearly; if the maximum occurs at an endpoint, the mode is that endpoint.

Trap: the mode is not the maximum value of f; it is the value of x at which the maximum occurs.

WORKED EXAMPLE 11 **Mode of a quadratic density**

Let X have density $f(x) = \frac{3}{4}x(2 - x)$ for $0 \le x \le 2$, 0 otherwise. Find the mode of X.

SOLUTION

On $[0, 2]$, $f(x) = \frac{3}{2}x - \frac{3}{4}x^2$, so

$$f'(x) = \frac{3}{2} - \frac{3}{2}x = \frac{3}{2}(1 - x).$$

$f'(x) = 0$ gives $x = 1$. Moreover $f'(x) > 0$ for $x < 1$ and $f'(x) < 0$ for $x > 1$, so f increases then decreases: $x = 1$ is a maximum. Since $f(0) = f(2) = 0 < f(1) = \frac{3}{4}$, the global maximum on the support is at $x = 1$.

$$\boxed{\text{Mode} = 1.}$$

(Here the density is symmetric about $x = 1$, so mode = mean = median = 1.)

7.5 Skill 5 — Obtaining and using the cumulative distribution function

METHOD **From PDF to CDF and back**

1. Definition: $F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt$.
2. Build F piece by piece: $F(x) = 0$ below the support, $F(x) = 1$ above it; on the support, integrate f from the left endpoint.
3. Check: F is continuous, non-decreasing, with $F(-\infty) = 0$ and $F(+\infty) = 1$.
4. Probabilities from F: $P(a \le X \le b) = F(b) - F(a)$; $P(X > a) = 1 - F(a)$.
5. PDF from CDF: $f(x) = F'(x)$ wherever F is differentiable.

WORKED EXAMPLE 12 **Building the CDF**

The random variable X has density

$$f(x) = \begin{cases} \frac{3x^2}{8}, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

1. Find the cumulative distribution function $F(x)$, specifying it for all real x.
2. Use F to compute $P(1 \le X \le 1.5)$.
3. The random variable Y has CDF $G(y) = \begin{cases} 0, & y < 0 \\ y^3, & 0 \le y \le 1 \\ 1, & y > 1 \end{cases}$. Find the density of Y.

SOLUTION

1. The CDF. For $x < 0$: $F(x) = 0$. For $0 \le x \le 2$:

$$F(x) = \int_0^x \frac{3t^2}{8}\,dt = \left[\frac{t^3}{8}\right]_0^x = \frac{x^3}{8}.$$

For $x > 2$: $F(x) = 1$. Hence

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{x^3}{8}, & 0 \le x \le 2, \\ 1, & x > 2. \end{cases}$$

Check: $F(0) = 0$, $F(2) = 1$, F continuous and increasing. ✓

2. Probability from F.

$$P(1 \le X \le 1.5) = F(1.5) - F(1) = \frac{1.5^3}{8} - \frac{1}{8} = \frac{3.375 - 1}{8} = \frac{2.375}{8} = 0.296875.$$

3. Density of Y. Differentiate G:

$$g(y) = G'(y) = \begin{cases} 3y^2, & 0 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Check: $\int_0^1 3y^2\,dy = 1$ and $g \ge 0$: g is a valid density.

7.6 Skill 6 — Median, quartiles and quantiles

METHOD **Solving $F(q) = p$**

1. The **median** m satisfies $F(m) = \frac{1}{2}$; the **lower quartile** Q_1 satisfies $F(Q_1) = \frac{1}{4}$; the **upper**

quartile Q_3 satisfies $F(Q_3) = \frac{3}{4}$.

2. More generally the p-quantile q_p solves $F(q_p) = p$.
3. Set up the equation using the correct piece of F, solve it, and check that the solution lies in the support (reject extraneous roots).
4. The interquartile range is $Q_3 - Q_1$.

Reflex: if the equation has several roots, only the root inside the support of X is acceptable.

WORKED EXAMPLE 13 **Median and quartiles**

The continuous random variable X has density

$$f(x) = \begin{cases} \frac{3}{64}x^2(4-x), & 0 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

1. Show that $F(x) = \frac{x^3}{64}\left(3 - \frac{3x}{4}\right) \cdot \frac{4}{3}$ is wrong as stated, and derive the correct CDF: $F(x) = \frac{x^3}{16} - \frac{3x^4}{256}$ on $[0, 4]$.
2. Verify that the median is $m = 2$.
3. Show that the lower quartile Q_1 satisfies $Q_1^4 - \frac{64}{3}Q_1^3 + \frac{1024}{3} \cdot \frac{1}{4} \cdot \frac{256}{256} = 0$, i.e. solve $F(Q_1) = \frac{1}{4}$ numerically to 2 d.p.

SOLUTION

1. The CDF. For $0 \le x \le 4$:

$$F(x) = \int_0^x \frac{3}{64}(4t^2 - t^3)\,dt = \frac{3}{64}\left[\frac{4t^3}{3} - \frac{t^4}{4}\right]_0^x = \frac{x^3}{16} - \frac{3x^4}{256}.$$

So $F(x) = 0$ for $x < 0$, $F(x) = \frac{x^3}{16} - \frac{3x^4}{256}$ for $0 \le x \le 4$, $F(x) = 1$ for $x > 4$. Check: $F(4) = \frac{64}{16} - \frac{3 \cdot 256}{256} = 4 - 3 = 1$. ✓

2. Median.

$$F(2) = \frac{8}{16} - \frac{3 \cdot 16}{256} = \frac{1}{2} - \frac{3}{16} = \frac{8-3}{16} = \frac{5}{16}.$$

This is not $\frac{1}{2}$, so $m \neq 2$; we solve $F(m) = \frac{1}{2}$:

$$\frac{m^3}{16} - \frac{3m^4}{256} = \frac{1}{2} \iff 3m^4 - 16m^3 + 128 = 0.$$

Testing $m = 2$: $48 - 128 + 128 = 48 \neq 0$. Testing $m \approx 2.4$: $3(33.1776) - 16(13.824) + 128 =$

$99.53 - 221.18 + 128 = 6.35$. Testing $m \approx 2.5$: $117.19 - 250 + 128 = -4.81$. Linear interpolation gives $m \approx 2.46$. Hence $m \approx 2.46$ (2 d.p.).
3. Lower quartile. Solve $F(Q_1) = \frac{1}{4}$:

$$\frac{Q_1^3}{16} - \frac{3Q_1^4}{256} = \frac{1}{4} \iff 3Q_1^4 - 16Q_1^3 + 64 = 0.$$

Try $Q_1 = 1.8$: $3(10.4976) - 16(5.832) + 64 = 31.49 - 93.31 + 64 = 2.18$. Try $Q_1 = 1.9$: $3(13.0321) - 16(6.859) + 64 = 39.10 - 109.74 + 64 = -6.65$. Interpolating: $Q_1 \approx 1.8 + 0.1 \times \frac{2.18}{2.18+6.65} \approx 1.82$. So $Q_1 \approx 1.82$ (2 d.p.).

7.7 Skill 7 — Full GCE-style synthesis problem

METHOD **Tackling a complete exam question**

1. Read fully and identify what is given (density with unknown constant, CDF, etc.).
2. Use $\int f = 1$ to find unknown constants.
3. Answer each part with the right tool: probabilities (integrate or use F), E, Var, mode (maximise f), median/quartiles (solve $F(q) = p$).
4. Present exact values, then decimals to the required accuracy.

WORKED EXAMPLE 14 **GCE-style synthesis**

The weekly demand X (in hundreds of litres) for petrol at a fuel station in Yaoundé is modelled by

$$f(x) = \begin{cases} kx(3-x)^2, & 0 \le x \le 3, \\ 0, & \text{otherwise.} \end{cases}$$

1. Show that $k = \frac{4}{27}$.
2. Find $E(X)$ and interpret it.
3. Find $\text{Var}(X)$.
4. Find the mode of X.
5. Find the CDF $F(x)$ and deduce the median, given that it satisfies $8m^4 - 36m^3 + \ldots$ — solve $F(m) = \frac{1}{2}$ to 2 d.p.
6. Find the probability that demand exceeds 200 litres in a week.

SOLUTION

1. Constant k. Expand: $x(3-x)^2 = x(9-6x+x^2) = 9x-6x^2+x^3$.

$$\int_0^3 k(9x-6x^2+x^3)\,dx = k\left[\frac{9x^2}{2} - 2x^3 + \frac{x^4}{4}\right]_0^3 = k\left(\frac{81}{2} - 54 + \frac{81}{4}\right) = k\cdot\frac{162-216+81}{4} = \frac{27k}{4}.$$

Setting $\frac{27k}{4} = 1$ gives $k = \frac{4}{27}$. Also $f(x) \geq 0$ on $[0,3]$, so f is a density.

2. Mean.

$$E(X) = \frac{4}{27}\int_0^3 x(9x-6x^2+x^3)\,dx = \frac{4}{27}\int_0^3 (9x^2-6x^3+x^4)\,dx$$

$$= \frac{4}{27}\left[3x^3 - \frac{3x^4}{2} + \frac{x^5}{5}\right]_0^3 = \frac{4}{27}\left(81 - \frac{243}{2} + \frac{243}{5}\right) = \frac{4}{27}\cdot\frac{810-1215+486}{10} = \frac{4}{27}\cdot\frac{81}{10} = \frac{6}{5} = 1.2.$$

Average weekly demand is about $1.2 \times 100 = 120$ litres.

3. Variance.

$$E(X^2) = \frac{4}{27}\int_0^3 (9x^3-6x^4+x^5)\,dx = \frac{4}{27}\left[\frac{9x^4}{4} - \frac{6x^5}{5} + \frac{x^6}{6}\right]_0^3$$

$$= \frac{4}{27}\left(\frac{729}{4} - \frac{1458}{5} + \frac{729}{6}\right) = \frac{4}{27}\cdot\frac{3645-5832+2430}{20} = \frac{4}{27}\cdot\frac{243}{20} = \frac{9}{5}.$$

$$\mathrm{Var}(X) = \frac{9}{5} - \left(\frac{6}{5}\right)^2 = \frac{9}{5} - \frac{36}{25} = \frac{45-36}{25} = \frac{9}{25} = 0.36.$$

4. Mode. $f(x) = \frac{4}{27}(9x-6x^2+x^3)$, so $f'(x) = \frac{4}{27}(9-12x+3x^2) = \frac{12}{27}(x^2-4x+3) = \frac{4}{9}(x-1)(x-3)$. On $[0,3]$: $f'(x) = 0$ at $x = 1$ and $x = 3$. Sign of f': positive on $[0,1)$, negative on $(1,3)$. So f has a maximum at $x = 1$. Since $f(0) = f(3) = 0$, the mode is $\boxed{1}$ (i.e. 100 litres).

5. CDF and median. For $0 \leq x \leq 3$:

$$F(x) = \frac{4}{27}\int_0^x (9t-6t^2+t^3)\,dt = \frac{4}{27}\left(\frac{9x^2}{2} - 2x^3 + \frac{x^4}{4}\right) = \frac{18x^2-8x^3+x^4}{27}\cdot\frac{2}{2} = \frac{2x^2(9)-8x^3+x^4}{27} = \frac{x^4-8x^3+18x^2}{27}.$$

Check: $F(3) = \frac{81-216+162}{27} = \frac{27}{27} = 1$. ✓

Median: solve $F(m) = \frac{1}{2}$, i.e. $m^4-8m^3+18m^2 = 13.5$. Try $m = 1.4$: $3.8416-21.952+35.28 = 17.17$ (too big). Try $m = 1.2$: $2.0736 - 13.824 + 25.92 = 14.17$. Try $m = 1.15$: $1.749 - 12.167 + 23.805 = 13.387$. Try $m = 1.16$: $1.8106-12.484+24.2208 = 13.547$. Interpolating between 1.15 and 1.16: $m \approx 1.15 + 0.01 \times \frac{13.5-13.387}{13.547-13.387} \approx 1.157$. So $m \approx 1.16$ (2 d.p.), i.e. about 116 litres.

6. $P(X > 2)$. Using the CDF:

$$P(X > 2) = 1 - F(2) = 1 - \frac{16-64+72}{27} = 1 - \frac{24}{27} = \frac{3}{27} = \frac{1}{9} \approx 0.111.$$

So there is about an 11% chance that weekly demand exceeds 200 litres.

WARNING **Common errors to avoid**

- Forgetting that f must be non-negative and integrate to 1 — both conditions are required.
- Writing $\mathrm{Var}(X) = E(X^2) - E(X)$ instead of $E(X^2) - [E(X)]^2$.
- Confusing the mode (value of x) with the maximum value of f.
- Solving $F(m) = \frac{1}{2}$ and accepting a root outside the support of X.
- Forgetting the pieces $F(x) = 0$ and $F(x) = 1$ when asked for the CDF "for all x".
- Believing $P(X = a) = f(a)$: for continuous variables, $P(X = a) = 0$; only areas give probabilities.

8 EXERCISES

8.1 I check my knowledge

EXERCISE 7 **MCQ: properties of a density function**

Let X be a continuous random variable with probability density function f. For each of the following statements, choose the single correct answer and justify briefly.

1. A function f is a probability density function on $\mathbb{R}$ if and only if:
   (a) $f(x) \geq 0$ for all x;
   (b) $\int_{-\infty}^{+\infty} f(x)\,dx = 1$;
   (c) $f(x) \geq 0$ for all x and $\int_{-\infty}^{+\infty} f(x)\,dx = 1$;
   (d) $0 \leq f(x) \leq 1$ for all x.
2. If X has density f, then $P(X = 2)$ equals:
   (a) $f(2)$;
   (b) 0;
   (c) $F(2)$;
   (d) $1 - F(2)$.
3. The cumulative distribution function F of a continuous random variable is always:
   (a) increasing and continuous;
   (b) decreasing and continuous;
   (c) increasing with possible jumps;
   (d) differentiable everywhere.
4. For a continuous random variable, the median m satisfies:
   (a) $f(m) = \frac{1}{2}$;
   (b) $F(m) = \frac{1}{2}$;
   (c) $E(X) = m$;
   (d) $P(X > m) = 1$.

EXERCISE 8 **True or false?**

Answer true or false to each statement, justifying your answer with a proof or a counter-example. X denotes a continuous random variable with density f and cumulative distribution function F.

1. If $f(x) > 1$ for some value of x, then f cannot be a probability density function.
2. $P(a \le X \le b) = P(a < X < b)$ for all real numbers $a < b$.
3. $E(X)$ always lies between the first quartile Q_1 and the third quartile Q_3.
4. For all $x \in \mathbb{R}$, $f(x) = F'(x)$ wherever F is differentiable.
5. The mode of X is the value of x at which F attains its maximum.

EXERCISE 9 **Definitions and vocabulary**

1. Define precisely: the **probability density function** of a continuous random variable; the **cumulative distribution function**; the **median**; the **mode**.
2. State the relation linking $P(X \le x)$, $P(X \ge x)$ and $F(x)$.
3. Write down the integral formulas giving $E(X)$ and $\mathrm{Var}(X)$ for a continuous random variable with density f on $\mathbb{R}$.
4. Give the equations that define the quartiles Q_1 and Q_3 of a continuous random variable.

EXERCISE 10 **Direct calculations**

Let X be the continuous random variable with density

$$f(x) = \begin{cases} 2x & \text{if } 0 \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

1. Verify that f is a probability density function.
2. Compute $P\left(\frac{1}{2} \le X \le \frac{3}{4}\right)$.
3. Determine the cumulative distribution function $F(x)$ for all $x \in \mathbb{R}$.
4. Compute $E(X)$.

8.2 I practise

EXERCISE 11 **A polynomial density on $[0,1]$**

Let X be the continuous random variable with density

$$f(x) = \begin{cases} 3x^2 & \text{if } 0 \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

1. Check that f is a density function and sketch its graph.
2. Compute $E(X)$ and $\mathrm{Var}(X)$.
3. Determine the mode of X.
4. Determine the cumulative distribution function F.
5. Compute the median m and the quartiles Q_1 and Q_3. Compare m with $E(X)$ and comment on the shape of the distribution.

EXERCISE 12 **Exponential waiting time**

The lifetime Y (in years) of an electronic component follows an exponential distribution with parameter $\lambda = 0.2$, that is,

$$f(y) = \begin{cases} 0.2\, e^{-0.2y} & \text{if } y \ge 0, \\ 0 & \text{if } y < 0. \end{cases}$$

1. Determine the cumulative distribution function F of Y.
2. Compute the probability that the component lasts more than 5 years.
3. Determine the median lifetime m.
4. Show that, for any exponential distribution, the probability that Y exceeds its median is exactly $\frac{1}{2}$, and compute this probability directly using the survival function $P(Y > y) = e^{-\lambda y}$.
5. Given that $E(Y) = \frac{1}{\lambda}$, compare the mean lifetime with the median lifetime.

EXERCISE 13 **Recovering the density from the CDF**

A continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & \text{if } x < 0, \\ \dfrac{x^2}{4} & \text{if } 0 \le x \le 2, \\ 1 & \text{if } x > 2. \end{cases}$$

1. Verify that F satisfies the properties of a cumulative distribution function.
2. Deduce the probability density function f of X.
3. Compute $E(X)$ and $\mathrm{Var}(X)$.
4. Determine the mode and the median of X.
5. Compute $P(1 \le X \le 1.5)$ in two different ways: using F, then using f.

EXERCISE 14 **A triangular distribution: river flow**

The daily flow D of a river (in $\mathrm{m^3\,s^{-1}}$) during the rainy season is modelled by a triangular distribution on $[10, 30]$ with mode at 20: the density f is affine on $[10, 20]$, affine on $[20, 30]$, zero outside $[10, 30]$, continuous, and maximal at $d = 20$.

1. Using the fact that the total area under the density curve equals 1, show that $f(20) = \dfrac{1}{10}$ and write down the expression of $f(d)$ on each interval.
2. Compute the probability that the flow lies between 15 and 25 $\mathrm{m^3\,s^{-1}}$.
3. Determine the cumulative distribution function F of D.
4. Determine the median flow and interpret the result for the management of a small hydroelectric dam on this river.

8.3 I prepare for the GCE

EXERCISE 15 **GCE style: a full study of a density**

(12 marks)

The continuous random variable X has probability density function

$$f(x) = \begin{cases} c(1 - x)^2 & \text{if } 0 \le x \le 1, \\ 0 & \text{otherwise,} \end{cases}$$

where c is a positive constant.

1. Show that $c = 3$. (2 marks)
2. Sketch the graph of f and state the mode of X. (2 marks)
3. Show that $E(X) = \frac{1}{4}$ and compute $\mathrm{Var}(X)$. (3 marks)
4. Find the cumulative distribution function $F(x)$ for all real x. (2 marks)
5. Show that the median m satisfies $(1 - m)^3 = \frac{1}{2}$, and deduce the exact value of m. (2 marks)
6. Without further calculation, explain why $m > E(X)$, relating your answer to the shape of the density. (1 mark)

EXERCISE 16 **GCE style: lifetime of a water pump**

(13 marks)

In a village of the North West Region, the time T (in months) before the first breakdown of a water pump is modelled as a continuous random variable with density

$$f(t) = \begin{cases} \dfrac{k}{t^2} & \text{if } t \geq 1, \\ 0 & \text{if } t < 1, \end{cases}$$

where k is a constant.

1. Show that $k = 1$. (2 marks)
2. Determine the cumulative distribution function F of T. (2 marks)
3. Compute the probability that the pump works without breakdown for more than 3 months. (2 marks)
4. Determine the median time before the first breakdown and interpret this value. (2 marks)
5. Determine the quartiles Q_1 and Q_3, and the interquartile range. (3 marks)
6. Show that $E(T)$ does not exist (the integral diverges). Explain why this model is nevertheless usable to compute probabilities and quantiles. (2 marks)

EXERCISE 17 **GCE style: synthesis problem on a density with parameter**

(15 marks)

A continuous random variable X, representing the mass (in kg) of a bag of cement leaving a factory, has density

$$f(x) = \begin{cases} a + bx & \text{if } 49 \le x \le 51, \\ 0 & \text{otherwise,} \end{cases}$$

where a and b are real constants. It is given that $E(X) = 50$.

1. Write down the two equations satisfied by a and b, and show that $a = \frac{1}{2}$ and $b = 0$. (4 marks)

2. Hence state the name of the distribution of X and write down its density. (1 mark)

3. Compute $\operatorname{Var}(X)$ and the standard deviation $\sigma(X)$. (3 marks)

4. Determine the cumulative distribution function F of X. (2 marks)

5. A bag is accepted if its mass lies between 49.5 kg and 50.5 kg. Compute the probability that a randomly chosen bag is accepted. (2 marks)

6. Determine the median and the quartiles of X. What do you notice? Justify your observation by a symmetry argument. (3 marks)

9 SOLUTIONS TO THE EXERCISES

10 EXERCICES CORRIGÉS — VARIABLES ALÉATOIRES CONTINUES

SOLUTION **Exercice 1 — QCM : propriétés d'une fonction de densité**

1. Réponse (c). Une fonction f est une densité de probabilité sur $\mathbb{R}$ si et seulement si les deux conditions suivantes sont satisfaites simultanément :

   - $f(x) \geq 0$ pour tout $x \in \mathbb{R}$ (positivité) ;
   - $\int_{-\infty}^{+\infty} f(x)\,dx = 1$ (probabilité totale égale à 1).

   La proposition (a) est insuffisante : $f(x) = 2$ sur $[0, 1]$ (et 0 ailleurs) est positive mais son intégrale vaut $2 \neq 1$. La proposition (b) est insuffisante : $f(x) = x$ sur $[-1, \sqrt{3}]$ (et 0 ailleurs) a une intégrale $\frac{(\sqrt{3})^2-(-1)^2}{2} = 1$, mais $f(x) < 0$ sur $[-1, 0)$. La proposition (d) est fausse : une densité n'est pas bornée par 1 (exemple : $f(x) = 2$ sur $[0, \frac{1}{2}]$).

2. Réponse (b). Pour une variable aléatoire continue, $P(X = 2) = \int_{2}^{2} f(x)\,dx = 0$: la surface sous la courbe au-dessus d'un point unique est nulle.

3. Réponse (a). $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$. Comme $f \geq 0$, F est croissante ; comme X est continue, F est continue sur $\mathbb{R}$ (pas de sauts). Elle n'est pas nécessairement dérivable partout (exemple : aux angles d'une densité par morceaux).

4. Réponse (b). La médiane m est définie par $F(m) = \frac{1}{2}$, c'est-à-dire $P(X \leq m) = P(X \geq m) = \frac{1}{2}$. L'affirmation (a) confond densité et probabilité ; (c) est fausse en général (moyenne ≠ médiane pour les lois asymétriques) ; (d) donnerait $P(X > m) = 1$, soit $F(m) = 0$.

SOLUTION **Exercice 2 — Vrai ou faux ?**

1. Faux. Une densité n'est pas tenue de satisfaire $f(x) \leq 1$; seules les probabilités sont bornées par 1, et les probabilités sont des aires sous f, non des valeurs de f. Contre-exemple : $f(x) = 2$ pour $x \in [0, \frac{1}{2}]$ et 0 ailleurs. Alors $f \geq 0$ et $\int_0^{1/2} 2\,dx = 1$, donc f est une densité, pourtant $f(x) = 2 > 1$.

2. Vrai. Comme X est continue, $P(X = a) = P(X = b) = 0$. Donc

$$P(a \le X \le b) = P(a < X < b) + P(X = a) + P(X = b) = P(a < X < b).$$

3. Faux. La moyenne n'est pas obligatoirement à l'intérieur de l'intervalle interquartile. Contre-exemple : soit X de densité

$$f(x) = \begin{cases} 0.99 & \text{si } 0 \le x \le 1, \\ 0.01 \times \frac{1}{10000} & \text{si } 1 < x \le 10001, \\ 0 & \text{sinon.} \end{cases}$$

(L'aire totale est $0.99 \times 1 + 0.01 \times 1 = 1$.) Sur $[0, 1]$, $F(x) = 0.99x$. Donc $Q_1 \approx 0.253$ et $Q_3 \approx 0.758$. La moyenne vaut $E(X) = 0.99 \times 0.5 + 0.01 \times \frac{10001+1}{2} = 0.495 + 50.01 = 50.505$, qui est strictement supérieure à Q_3. L'affirmation est donc fausse.

4. Vrai. D'après le théorème fondamental de l'analyse, puisque $F(x) = \int_{-\infty}^{x} f(t)\,dt$, on a $F'(x) = f(x)$ en tout point où f est continue, donc en tout point où F est dérivable.

5. Faux. Le mode est la valeur de x où la densité f atteint son maximum, pas F. Comme F est croissante avec $\lim_{x \to +\infty} F(x) = 1$, F n'a pas de maximum atteint en un point fini. Contre-exemple : pour $f(x) = 2x$ sur $[0, 1]$, le mode est $x = 1$ (maximum de f), alors que $F(x) = x^2$ croît strictement vers 1.

SOLUTION **Exercice 3 — Définitions et vocabulaire**

1. Fonction de densité de probabilité : fonction $f : \mathbb{R} \to \mathbb{R}$ telle que $f(x) \ge 0$ pour tout x et $\int_{-\infty}^{+\infty} f(x)\,dx = 1$; les probabilités se calculent par $P(a \le X \le b) = \int_a^b f(x)\,dx$. Fonction de répartition : fonction $F : \mathbb{R} \to [0, 1]$ définie par $F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt$. Médiane : réel m tel que $F(m) = \frac{1}{2}$, c'est-à-dire $P(X \le m) = P(X \ge m) = \frac{1}{2}$. Mode : valeur x_0 où la densité f atteint son maximum : $f(x_0) = \max f$.

2. $P(X \le x) = F(x)$ et, comme $P(X = x) = 0$, $P(X \ge x) = 1 - F(x)$. Donc $P(X \le x) + P(X \ge x) = 1$.

3. $E(X) = \int_{-\infty}^{+\infty} x f(x)\,dx$ et $\mathrm{Var}(X) = \int_{-\infty}^{+\infty} \big(x - E(X)\big)^2 f(x)\,dx = \int_{-\infty}^{+\infty} x^2 f(x)\,dx - \big(E(X)\big)^2$, pourvu que ces intégrales convergent.

4. Q_1 et Q_3 sont définis par $F(Q_1) = \frac{1}{4}$ et $F(Q_3) = \frac{3}{4}$, c'est-à-dire $\int_{-\infty}^{Q_1} f(x)\,dx = \frac{1}{4}$ et $\int_{-\infty}^{Q_3} f(x)\,dx = \frac{3}{4}$.

SOLUTION **Exercice 4 — Calculs directs : $f(x) = 2x$ sur $[0,1]$**

1. $f(x) = 2x \geq 0$ sur $[0,1]$ et $f = 0$ ailleurs, donc $f \geq 0$ sur $\mathbb{R}$. De plus
$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_0^1 2x\,dx = \left[x^2\right]_0^1 = 1.$$
Les deux conditions sont vérifiées, f est bien une densité de probabilité.

2. $P\left(\frac{1}{2} \leq X \leq \frac{3}{4}\right) = \int_{1/2}^{3/4} 2x\,dx = \left[x^2\right]_{1/2}^{3/4} = \frac{9}{16} - \frac{1}{4} = \frac{5}{16} = 0.3125.$

3. Pour $x < 0$: $F(x) = 0$. Pour $0 \leq x \leq 1$: $F(x) = \int_0^x 2t\,dt = x^2$. Pour $x > 1$: $F(x) = 1$. Donc
$$F(x) = \begin{cases} 0 & \text{si } x < 0, \\ x^2 & \text{si } 0 \leq x \leq 1, \\ 1 & \text{si } x > 1. \end{cases}$$

4. $E(X) = \int_0^1 x \cdot 2x\,dx = \int_0^1 2x^2\,dx = \left[\frac{2x^3}{3}\right]_0^1 = \frac{2}{3}.$

SOLUTION **Exercice 5 — Densité polynomiale sur $[0,1]$: $f(x) = 3x^2$**

1. $f(x) = 3x^2 \geq 0$ sur $[0,1]$, $f = 0$ ailleurs, et
$$\int_0^1 3x^2\,dx = \left[x^3\right]_0^1 = 1.$$
f est une densité. Son graphe est l'arc de parabole $y = 3x^2$ sur $[0,1]$, partant de $(0,0)$ pour arriver à $(1,3)$, et l'axe des abscisses hors de $[0,1]$.

2. $E(X) = \int_0^1 x \cdot 3x^2\,dx = \int_0^1 3x^3\,dx = \frac{3}{4}$. $E(X^2) = \int_0^1 x^2 \cdot 3x^2\,dx = \int_0^1 3x^4\,dx = \frac{3}{5}$.
$$\mathrm{Var}(X) = E(X^2) - \left(E(X)\right)^2 = \frac{3}{5} - \frac{9}{16} = \frac{48-45}{80} = \frac{3}{80} = 0.0375.$$

3. f est strictement croissante sur $[0,1]$ (car $f'(x) = 6x > 0$ pour $x > 0$), son maximum sur le support est atteint en $x = 1$. Le **mode** est 1.

4. Pour $x < 0$: $F(x) = 0$. Pour $0 \leq x \leq 1$: $F(x) = \int_0^x 3t^2\,dt = x^3$. Pour $x > 1$: $F(x) = 1$.

5. Médiane : $F(m) = m^3 = \frac{1}{2} \Rightarrow m = \sqrt[3]{\frac{1}{2}} = 2^{-1/3} \approx 0.794$. Quartiles : $Q_1^3 = \frac{1}{4} \Rightarrow Q_1 = 4^{-1/3} \approx 0.630$; $Q_3^3 = \frac{3}{4} \Rightarrow Q_3 = \left(\frac{3}{4}\right)^{1/3} \approx 0.909$. Comparaison : $E(X) = 0.75 < m \approx$

0.794. La densité croît vers 1 (masse concentrée près de la borne droite, queue étirée à gauche) : la distribution est asymétrique à gauche (asymétrie négative), ce qui tire la moyenne en dessous de la médiane. Ceci est cohérent avec l'ordre mode $1 > m > E(X)$.

SOLUTION **Exercice 6 — Temps d'attente exponentiel ($\lambda = 0.2$)**

1. Pour $y < 0$: $F(y) = 0$. Pour $y \geq 0$:

$$F(y) = \int_0^y 0.2\, e^{-0.2t}\, dt = \Big[- e^{-0.2t} \Big]_0^y = 1 - e^{-0.2y}.$$

2. $P(Y > 5) = 1 - F(5) = e^{-0.2\times 5} = e^{-1} \approx 0.368$.

3. Médiane : $F(m) = \frac{1}{2} \Leftrightarrow 1 - e^{-0.2m} = \frac{1}{2} \Leftrightarrow e^{-0.2m} = \frac{1}{2} \Leftrightarrow 0.2m = \ln 2$, donc

$$m = \frac{\ln 2}{0.2} = 5 \ln 2 \approx 3.47 \text{ ans.}$$

4. Par définition de la médiane, $P(Y > m) = 1 - F(m) = 1 - \frac{1}{2} = \frac{1}{2}$. Directement avec la fonction de survie :

$$P(Y > m) = e^{-\lambda m} = e^{-\lambda \cdot \frac{\ln 2}{\lambda}} = e^{-\ln 2} = \frac{1}{2}.$$

5. $E(Y) = \frac{1}{\lambda} = \frac{1}{0.2} = 5$ ans, alors que $m = 5 \ln 2 \approx 3.47$ ans. Donc $E(Y) > m$: la loi exponentielle est asymétrique à droite (longue queue de très grandes durées de vie), ce qui tire la moyenne au-dessus de la médiane. Concrètement : la moitié des composants tombent en panne avant environ 3, 5 ans, mais la durée de vie moyenne est de 5 ans.

SOLUTION **Exercice 7 — Retrouver la densité à partir de la fonction de répartition**

1. Vérification des trois propriétés d'une fonction de répartition :
   - Limites : $F(x) = 0$ pour $x < 0$ donc $\lim_{-\infty} F = 0$; $F(x) = 1$ pour $x > 2$ donc $\lim_{+\infty} F = 1$.
   - Croissance : sur $[0, 2]$, $F(x) = \frac{x^2}{4}$ a pour dérivée $\frac{x}{2} \geq 0$; constante ailleurs. Donc F est croissante sur $\mathbb{R}$.
   - Continuité : en $x = 0$: $F(0^-) = 0 = F(0)$; en $x = 2$: $F(2) = \frac{4}{4} = 1 = F(2^+)$. F est continue partout.

   F est bien une fonction de répartition.

2. $f(x) = F'(x)$ là où F est dérivable : $f(x) = \frac{x}{2}$ pour $0 \leq x \leq 2$, et $f(x) = 0$ sinon. (Vérification :

$\int_0^2 \frac{x}{2}\,dx = \left[\frac{x^2}{4}\right]_0^2 = 1$. ✓)

3. $E(X) = \int_0^2 x \cdot \frac{x}{2}\,dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{8}{6} = \frac{4}{3}$. $E(X^2) = \int_0^2 x^2 \cdot \frac{x}{2}\,dx = \left[\frac{x^4}{8}\right]_0^2 = \frac{16}{8} = 2$.

$$\mathrm{Var}(X) = 2 - \left(\frac{4}{3}\right)^2 = 2 - \frac{16}{9} = \frac{2}{9} \approx 0.222.$$

4. Mode : $f(x) = \frac{x}{2}$ est croissante sur $[0,2]$, maximale en $x = 2$: le mode est 2. Médiane : $F(m) = \frac{1}{2} \Leftrightarrow \frac{m^2}{4} = \frac{1}{2} \Leftrightarrow m^2 = 2 \Leftrightarrow m = \sqrt{2} \approx 1.414$ (on prend la racine dans $[0,2]$).

5. Par F : $P(1 \le X \le 1.5) = F(1.5) - F(1) = \frac{1.5^2}{4} - \frac{1}{4} = \frac{2.25 - 1}{4} = \frac{1.25}{4} = 0.3125$. Par f : $P(1 \le X \le 1.5) = \int_1^{1.5} \frac{x}{2}\,dx = \left[\frac{x^2}{4}\right]_1^{1.5} = \frac{2.25 - 1}{4} = 0.3125$. Même résultat, comme attendu.

SOLUTION **Exercice 8 — Distribution triangulaire : débit d'une rivière**

1. Le graphe de f sur $[10, 30]$ est un triangle de base $30 - 10 = 20$ et de hauteur $h = f(20)$. L'aire totale vaut 1 :

$$\frac{1}{2} \times 20 \times h = 1 \quad \Longrightarrow \quad h = \frac{1}{10}.$$

Sur $[10, 20]$, la droite passant par $(10, 0)$ et $(20, \frac{1}{10})$ a pour pente $\frac{1/10}{10} = \frac{1}{100}$: $f(d) = \frac{d - 10}{100}$. Sur $[20, 30]$, la droite passant par $(20, \frac{1}{10})$ et $(30, 0)$ a pour pente $-\frac{1}{100}$: $f(d) = \frac{30 - d}{100}$. Ainsi

$$f(d) = \begin{cases} \frac{d - 10}{100} & \text{si } 10 \le d \le 20, \\ \frac{30 - d}{100} & \text{si } 20 \le d \le 30, \\ 0 & \text{sinon.} \end{cases}$$

2. Par symétrie du triangle par rapport à $d = 20$, l'intervalle $[15, 25]$ est symétrique par rapport au mode. Calcul :

$$P(15 \le D \le 25) = 2\int_{15}^{20} \frac{d - 10}{100}\,dd = 2\left[\frac{(d - 10)^2}{200}\right]_{15}^{20} = 2 \times \frac{100 - 25}{200} = \frac{150}{200} = 0.75.$$

(Autrement : aire totale 1 moins les deux triangles extérieurs d'aire $\frac{1}{2} \times 5 \times \frac{5}{100} = 0.125$ chacun : $1 - 0.25 = 0.75$.)

3. Pour $d < 10$: $F(d) = 0$. Pour $10 \le d \le 20$:

$$F(d) = \int_{10}^{d} \frac{t-10}{100}\,dt = \frac{(d-10)^2}{200}.$$

Pour $20 \le d \le 30$: $F(d) = F(20) + \int_{20}^{d} \frac{30-t}{100}\,dt = \frac{1}{2} + \left[\frac{30t - t^2/2}{100}\right]_{20}^{d}$. En calculant : $\int_{20}^{d} \frac{30-t}{100}dt = \frac{1}{100}\left[30(d-20) - \frac{d^2-400}{2}\right] = \frac{-d^2+60d-800}{200}$. Donc

$$F(d) = \frac{-d^2+60d-800}{200} + \frac{1}{2} = \frac{-d^2+60d-700}{200} = 1 - \frac{(30-d)^2}{200}.$$

Pour $d > 30$: $F(d) = 1$. (Vérification en $d = 20$: $1 - \frac{100}{200} = \frac{1}{2}$ ✓ ; en $d = 30$: 1 ✓.)

4. Médiane : par symétrie de la densité par rapport à $d = 20$, $F(20) = \frac{1}{2}$, donc $m = 20\ \text{m}^3\,\text{s}^{-1}$. (Algébriquement, sur $[10, 20]$: $\frac{(m-10)^2}{200} = \frac{1}{2} \Rightarrow (m-10)^2 = 100 \Rightarrow m = 20$.) Interprétation : la moitié des jours de la saison des pluies, le débit du fleuve est inférieur à $20\ \text{m}^3\,\text{s}^{-1}$, et l'autre moitié supérieur. Pour le gestionnaire du barrage, une turbine dimensionnée pour un débit de $20\ \text{m}^3\,\text{s}^{-1}$ tournera en dessous de sa capacité un jour sur deux en moyenne ; la médiane — et non le mode seul — est la valeur de référence pour le dimensionnement en capacité garantie.

SOLUTION **Exercice 9 — Type GCE : étude complète d'une densité $c(1-x)^2$ sur $[0,1]$**

1. f est une densité, donc $\int_0^1 c(1-x)^2\,dx = 1$. Par le changement de variable $u = 1 - x$:

$$\int_0^1 c(1-x)^2\,dx = c\left[-\frac{(1-x)^3}{3}\right]_0^1 = c\left(0 + \frac{1}{3}\right) = \frac{c}{3} = 1 \quad \Longrightarrow \quad c = 3.$$

[2 points : 1 pour l'intégrale égale à 1, 1 pour la valeur correcte.]

2. $f(x) = 3(1-x)^2$ sur $[0,1]$: arc de parabole décroissant de $f(0) = 3$ à $f(1) = 0$, nul hors $[0,1]$. Comme $f'(x) = -6(1-x) \le 0$ sur $[0,1]$, f est décroissante et maximale en $x = 0$: le **mode** est 0. [2 points : 1 pour un croquis correct (courbe passant par $(0,3)$ et $(1,0)$), 1 pour le mode.]

3. $E(X) = \int_0^1 3x(1-x)^2\,dx = 3\int_0^1 (x - 2x^2 + x^3)\,dx = 3\left[\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right] = 3 \times \frac{6-8+3}{12} = 3 \times \frac{1}{12} = \frac{1}{4}$.
$E(X^2) = \int_0^1 3x^2(1-x)^2\,dx = 3\int_0^1 (x^2 - 2x^3 + x^4)\,dx = 3\left[\frac{1}{3} - \frac{1}{2} + \frac{1}{5}\right] = 3 \times \frac{10-15+6}{30} = \frac{1}{10}$.

$$\text{Var}(X) = \frac{1}{10} - \left(\frac{1}{4}\right)^2 = \frac{1}{10} - \frac{1}{16} = \frac{8-5}{80} = \frac{3}{80} = 0.0375.$$

[3 points : 1 pour $E(X) = \frac{1}{4}$ montré, 1 pour $E(X^2)$, 1 pour la variance.]

4. Pour $x < 0$: $F(x) = 0$. Pour $0 \le x \le 1$:

$$F(x) = \int_0^x 3(1-t)^2\,dt = \Big[-(1-t)^3\Big]_0^x = 1-(1-x)^3.$$

Pour $x > 1$: $F(x) = 1$. [2 points : 1 pour l'expression intégrale, 1 pour l'énoncé complet en trois parties.]

5. $F(m) = \frac{1}{2} \Leftrightarrow 1-(1-m)^3 = \frac{1}{2} \Leftrightarrow (1-m)^3 = \frac{1}{2}$. Donc $1-m = \left(\frac{1}{2}\right)^{1/3} = 2^{-1/3}$, soit

$$m = 1 - \frac{1}{\sqrt[3]{2}} = 1 - 2^{-1/3} \approx 0.206.$$

[2 points : 1 pour l'équation, 1 pour la valeur exacte.]

6. La densité est décroissante sur $[0,1]$: la masse est concentrée près de 0 (les petites valeurs de X sont les plus probables) avec une queue étirée vers 1. Cette asymétrie à droite tire la moyenne au-dessus du gros des données, d'où $E(X) = 0.25 > m \approx 0.206$. [1 point pour un commentaire correct liant l'asymétrie à la position de la moyenne par rapport à la médiane.]

Attendus de l'examinateur : utilisation claire de $\int f = 1$; développement de $(1-x)^2$ avant intégration ; la fonction de répartition donnée sur tout $\mathbb{R}$; réponses exactes laissées sous forme de radicaux.

SOLUTION **Exercice 10 — Type GCE : durée de vie d'une pompe à eau**

1. $\int_1^{+\infty} \frac{k}{t^2}\,dt = k\left[-\frac{1}{t}\right]_1^{+\infty} = k(0+1) = k = 1$, car la probabilité totale doit valoir 1. Donc $k = 1$. [2 points : 1 pour l'intégrale impropre correctement écrite avec une limite, 1 pour la valeur.]

2. Pour $t < 1$: $F(t) = 0$. Pour $t \ge 1$:

$$F(t) = \int_1^t \frac{du}{u^2} = \left[-\frac{1}{u}\right]_1^t = 1 - \frac{1}{t}.$$

[2 points.]

3. $P(T > 3) = 1 - F(3) = \frac{1}{3} \approx 0.333$. La probabilité que la pompe fonctionne plus de 3 mois sans panne est $\frac{1}{3}$. [2 points.]

4. Médiane : $F(m) = \frac{1}{2} \Leftrightarrow 1 - \frac{1}{m} = \frac{1}{2} \Leftrightarrow \frac{1}{m} = \frac{1}{2} \Leftrightarrow m = 2$ mois. Interprétation : la moitié des pompes subissent leur première panne avant 2 mois de service, l'autre moitié après.

Une visite de maintenance devrait donc être programmée bien avant 2 mois si l'on veut prévenir les pannes pour la majorité des pompes. [2 points : 1 pour $m = 2$, 1 pour l'interprétation.]

5. $Q_1 : 1 - \frac{1}{Q_1} = \frac{1}{4} \Rightarrow \frac{1}{Q_1} = \frac{3}{4} \Rightarrow Q_1 = \frac{4}{3} \approx 1.33$ mois. $Q_3 : 1 - \frac{1}{Q_3} = \frac{3}{4} \Rightarrow \frac{1}{Q_3} = \frac{1}{4} \Rightarrow Q_3 = 4$ mois. Écart interquartile : $Q_3 - Q_1 = 4 - \frac{4}{3} = \frac{8}{3} \approx 2.67$ mois. [3 points : 1 pour chaque valeur.]

6. $E(T) = \int_1^{+\infty} t \cdot \frac{1}{t^2}\,dt = \int_1^{+\infty} \frac{dt}{t} = \left[\ln t\right]_1^{+\infty} = +\infty$: l'intégrale diverge, donc $E(T)$ n'existe pas (la loi a une queue trop lourde). Cependant, le modèle reste parfaitement utilisable pour les probabilités et les quantiles : ceux-ci ne requièrent que la fonction de répartition $F(t) = 1 - \frac{1}{t}$, qui est bien définie, puisque $\int_1^{+\infty} f = 1$ converge et F est une fonction de répartition correcte. L'inexistence de la moyenne signifie simplement que « durée de vie moyenne » n'est pas un résumé pertinent pour ce modèle ; il faut utiliser les médianes et les quartiles. [2 points : 1 pour la divergence montrée, 1 pour le commentaire.]

Attendus de l'examinateur : intégrales impropres gérées avec une limite $A \to +\infty$; valeurs exactes ; phrase d'interprétation claire dans le contexte (pompe du village) pour la médiane.

SOLUTION **Exercice 11 — Type GCE : problème de synthèse sur une densité à paramètre**

1. Première équation (probabilité totale) :

$$\int_{49}^{51} (a + bx)\,dx = \left[ax + \tfrac{b}{2}x^2\right]_{49}^{51} = 2a + \frac{b}{2}(51^2 - 49^2) = 2a + \frac{b}{2}(2601 - 2401) = 2a + 100b = 1.$$

Deuxième équation ($E(X) = 50$) :

$$\int_{49}^{51} x(a+bx)\,dx = \left[\tfrac{a}{2}x^2 + \tfrac{b}{3}x^3\right]_{49}^{51} = \frac{a}{2}(200) + \frac{b}{3}(51^3 - 49^3) = 100a + \frac{b}{3}(132651 - 117649) = 100a + \frac{15002}{3}b = 50.$$

D'après la première équation, $a = \frac{1 - 100b}{2}$. En remplaçant dans la seconde :

$$100 \times \frac{1 - 100b}{2} + \frac{15002}{3}b = 50 \implies 50 - 5000b + \frac{15002}{3}b = 50 \implies b\left(\frac{15002}{3} - 5000\right) = 0 \implies \frac{2}{3}b = 0,$$

donc $b = 0$ et alors $a = \frac{1}{2}$. [4 points : 1 par équation correctement posée, 1 pour la résolution, 1 pour les valeurs.] (Argument plus rapide accepté : l'intervalle $[49, 51]$ est symétrique par rapport à 50 ; $E(X) = 50$ force le terme linéaire bx à s'annuler, et $\int_{49}^{51} a\,dx = 2a = 1$ donne $a = \frac{1}{2}$.)

2. Avec $a = \frac{1}{2}$, $b = 0$: $f(x) = \frac{1}{2}$ sur $[49, 51]$, 0 ailleurs. C'est la **loi uniforme** sur $[49, 51]$, notée $X \sim U([49, 51])$. [1 point.]

3. $E(X^2) = \int_{49}^{51} \frac{x^2}{2}\,dx = \frac{51^3 - 49^3}{6} = \frac{15002}{6} = \frac{7501}{3}$.

$$\mathrm{Var}(X) = \frac{7501}{3} - 2500 = \frac{7501 - 7500}{3} = \frac{1}{3} \approx 0.333, \qquad \sigma(X) = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}} \approx 0.577\ \text{kg}.$$

(Ceci coïncide avec la formule standard $\mathrm{Var} = \frac{(b-a)^2}{12} = \frac{4}{12} = \frac{1}{3}$.) [3 points : 1 pour $E(X^2)$, 1 pour la variance, 1 pour σ.]

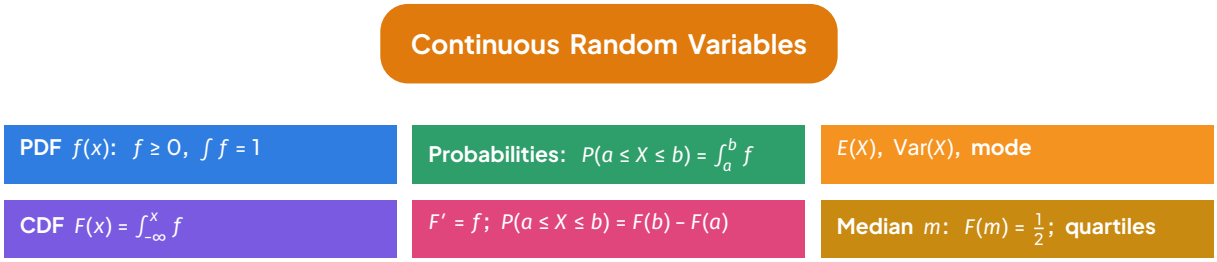
4. Pour $x < 49$: $F(x) = 0$. Pour $49 \le x \le 51$: $F(x) = \int_{49}^{x} \frac{dt}{2} = \frac{x - 49}{2}$. Pour $x > 51$: $F(x) = 1$. [2 points.]

5. $P(49.5 \le X \le 50.5) = F(50.5) - F(49.5) = \frac{1.5}{2} - \frac{0.5}{2} = \frac{1}{2}$. La moitié des sacs sont acceptés. [2 points.]

6. Médiane : $F(m) = \frac{1}{2} \Rightarrow \frac{m - 49}{2} = \frac{1}{2} \Rightarrow m = 50$ kg. Quartiles : $\frac{Q_1 - 49}{2} = \frac{1}{4} \Rightarrow Q_1 = 49.5$ kg ; $\frac{Q_3 - 49}{2} = \frac{3}{4} \Rightarrow Q_3 = 50.5$ kg. Observation : $E(X) = m = 50$ et les quartiles sont symétriques par rapport à 50 ($Q_1 = 50 - 0.5$, $Q_3 = 50 + 0.5$). Justification : la densité est constante, donc symétrique par rapport au centre $x = 50$ de l'intervalle : $f(50 + h) = f(50 - h)$ pour tout h. Pour toute densité symétrique, la moyenne et la médiane coïncident avec le centre de symétrie, et les quartiles sont à distances égales de ce centre. [3 points : 1 pour les trois valeurs, 1 pour l'observation, 1 pour l'argument de symétrie.]

Attendus de l'examinateur : calculs exacts avec $51^2 - 49^2 = 200$ et $51^3 - 49^3 = 15002$ (les identités remarquables différence de carrés/cubes sont les bienvenues) ; reconnaissance de la loi uniforme ; argument de symétrie correctement articulé à la dernière question.

11 SUMMARY SHEET



Mind map of Chapter FMA19: Continuous Random Variables.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- A **continuous random variable** X takes any value in an interval of $\mathbb{R}$; probabilities are given by areas under a curve.
- A **probability density function** (PDF) f satisfies: $f(x) \geq 0$ for all x, and $\displaystyle\int_{-\infty}^{+\infty} f(x)\,dx = 1$.
- $P(a \leq X \leq b) = \displaystyle\int_a^b f(x)\,dx$. For a continuous variable, $P(X = c) = 0$ for every single value c.
- The **cumulative distribution function** (CDF): $F(x) = P(X \leq x) = \displaystyle\int_{-\infty}^{x} f(t)\,dt$.
- The **mode** is the value of x where f is maximum; the **median** m satisfies $F(m) = \frac{1}{2}$; **quartiles** Q_1, Q_3 satisfy $F(Q_1) = \frac{1}{4}$, $F(Q_3) = \frac{3}{4}$.

Formulas and results to know

- Expectation: $E(X) = \displaystyle\int_{-\infty}^{+\infty} x\,f(x)\,dx$; $\quad E(g(X)) = \displaystyle\int_{-\infty}^{+\infty} g(x)f(x)\,dx$.
- Variance: $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$ with $E(X^2) = \displaystyle\int x^2 f(x)\,dx$.
- Linearity: $E(aX + b) = aE(X) + b$; $\quad \mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$.
- Properties of the CDF: F is increasing and continuous; $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to+\infty} F(x) = 1$; $F'(x) = f(x)$ where f is continuous.
- $P(a \leq X \leq b) = F(b) - F(a)$; $\quad P(X \geq a) = 1 - F(a)$.
- Unknown constant k in a PDF: solve $\displaystyle\int f(x)\,dx = 1$.

Methods

1. Find a constant k: write the total-area condition over the support only, integrate, solve.
2. Build F piecewise: $F(x) = 0$ below the support, $F(x) = 1$ above it, and $F(x) = \int_{\text{lower bound}}^{x} f(t)\,dt$ inside; check continuity at the joints.
3. Median/quartiles: solve $F(x) = p$ in the correct piece; reject roots outside the interval.
4. Mode: differentiate f on its support, solve $f'(x) = 0$, confirm a maximum (sign of f' or $f'' < 0$); check endpoints too.

Common mistakes

- Believing $f(x)$ is a probability: $f(x)$ can exceed 1; only areas are probabilities.
- Writing $P(X = c) > 0$: for a continuous variable $P(X = c) = 0$, so $P(a \le X \le b) = P(a < X < b)$.
- Forgetting the pieces of F outside the support, or getting a discontinuous F.
- Mixing up $\text{Var}(X) = E(X^2) - (E(X))^2$ with $(E(X))^2 - E(X^2)$ (variance is non-negative).
- Solving $F(x) = \frac{1}{2}$ in the wrong piece of a piecewise CDF.
- Integrating over $\mathbb{R}$ when the support is a bounded interval.

GCE tips

- Always state the support of X first; it fixes all integration bounds.
- Show the condition $\int f = 1$ explicitly when justifying that f is a PDF.
- Check your CDF: it must be continuous, increasing, from 0 to 1 — a quick sanity check earns method marks.
- For quartiles, write the equation $F(Q) = p$ before solving; examiners reward the setup.
- Keep exact values (fractions, surds) until the final line, then round to the required accuracy.

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